
AI & CAPITAL MARKETS

How LLMs Became a Capital Markets Story

How large language models work, why better models need vastly more capital, and how debt and equity markets — not labs — now shape the buildout.

Ashleyn Castelino · September 2026 · Educational purposes only — not investment advice

Where we're going

01 Models, chips and cost

Next-token prediction, training vs. inference, custom silicon

02 Why costs exploded

Scaling laws turned progress into capex

03 Debt & equity step in

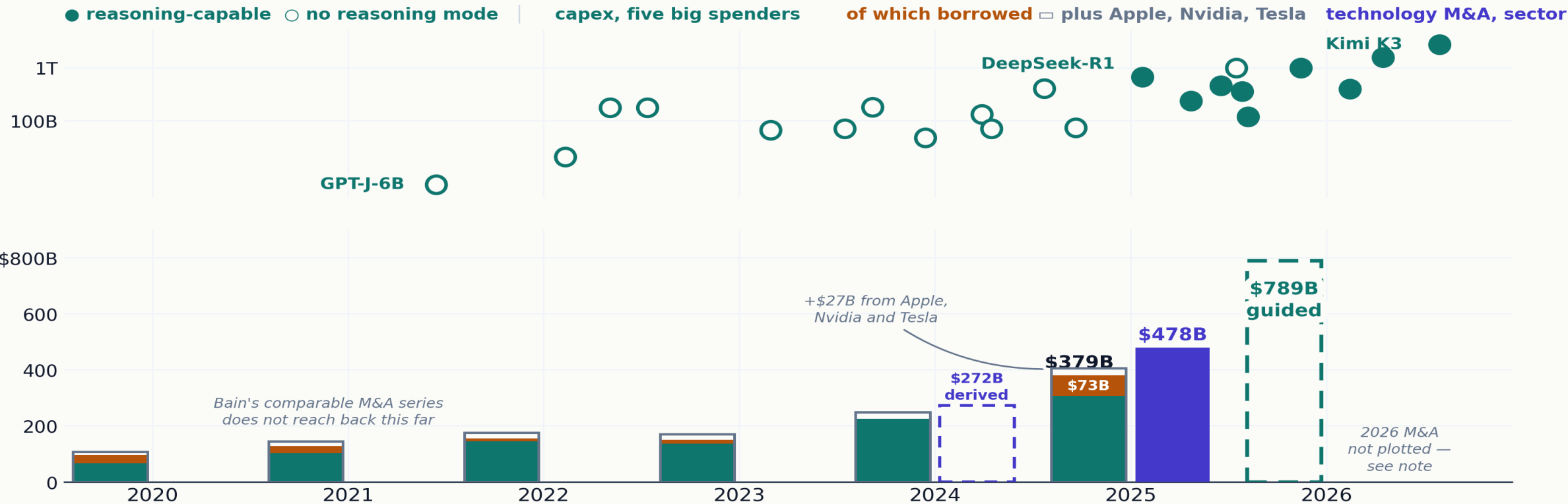
Bonds, private credit, and mega-rounds

04 Science to business case

Revenue, unit economics, and the fine print

The models got bigger, and the capital bill went vertical

Capability above, capital below. The solid bars are the five AI infrastructure spenders; the outline adds the rest of the Magnificent 7.



Model sizes — developers' own model cards and technical reports: GPT-J-6B (EleutherAI, June 2021), GPT-NeoX-20B (weights 9-Feb-2022), OPT-175B (arXiv 2205.01068), BLOOM (arXiv 2211.05100, July 2022), Llama 65B, Llama 2 70B, Falcon 180B, DBRX (132B total, 36B active, Databricks, 27-Mar-2024), Llama 3 70B, Llama 3.1 405B, Qwen2.5 72B, MiniMax-M1 (456B total, 45.9B active, arXiv 2506.13585), GLM-4.5 (355B total, 32B active, arXiv 2508.06471) and the 2025–26 frontier points. Counts are total parameters, not active, so mixture-of-experts models read larger than what runs per token. Only open-weight labs publish parameter counts, so the closed frontier — GPT-5.x, Claude, Gemini, Grok — does not appear; reasoning status is the developer's own; the model axis is a log scale. Capex — SEC 10-K cash flow statements, 2020 to 2025, excluding finance leases. Solid bars: Microsoft, Alphabet, Amazon, Meta and Oracle. The slate outline adds Apple, Nvidia and Tesla, for the full Magnificent 7 plus Oracle; Apple's line is PP&E, Nvidia's also covers intangibles. Nvidia and Oracle sit in the calendar year their fiscal year mostly covers, Microsoft on its June fiscal year as reported. Nvidia's year to January 2022 is not tagged in SEC XBRL, so the 2021 outline excludes it. Net new debt is proceeds less repayments, negative in 2024. The 2026 bar is dashed and covers only the five: it sums Amazon, Alphabet, Meta and Microsoft's own 2026 guidance (Aug 2026, \$733B) with Oracle's FY2026 capex of \$55.7B. M&A — Bain & Company, January 2026: tech deal value rose over 76% to \$478B year-to-date, and almost half of strategic tech deal value above \$500M involved AI natives or cited AI benefits; the 2024 bar is derived from that rate. No 2026 M&A bar: LSEG's \$649B for H1 2026 differs.

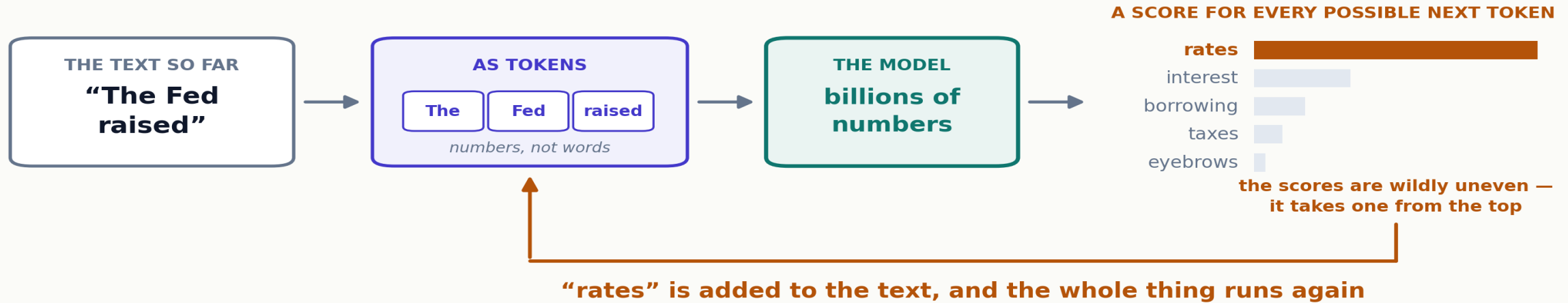
01

Models, chips and the cost of a token

You don't need the math — just enough mechanics to see where the money goes, and why both labs are now designing their own chips.

An LLM only ever does one thing: guess the next token

Writing, reasoning, coding — all of it is this one operation, run again and again.



That loop is the entire product.

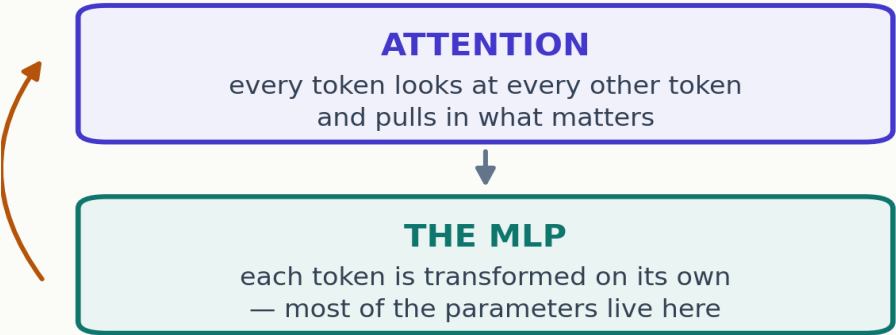
Doing it once, very fast, for millions of people, is the entire cost.
Everything in this deck follows from those two sentences.

The candidate words and the relative bar lengths are illustrative: they show that the model scores every token in its vocabulary and that the scores are very uneven, not any measured output. No probability figures are shown, because none were measured for this chart. Token boundaries also vary by tokenizer — the split shown is indicative.

Two operations, stacked a hundred times — and teaching them costs 3x running them

The architecture on the left is simple. The loop on the right is what makes training a capital project.

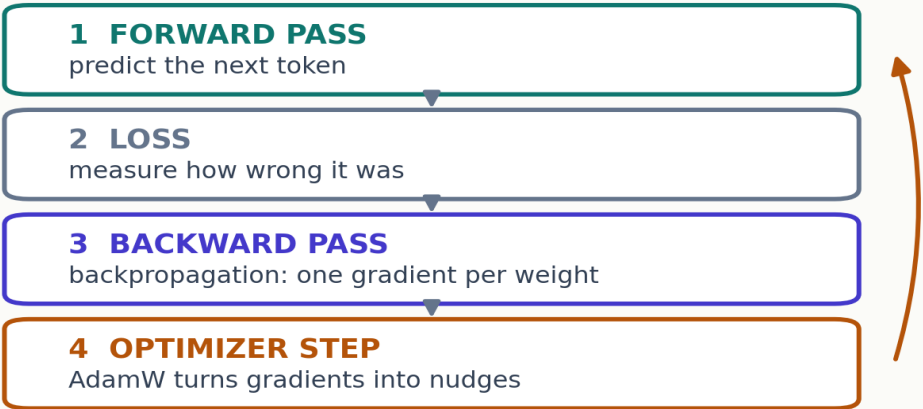
ONE TRANSFORMER LAYER



× 80 to 120 layers

Attention is why reading a prompt is one parallel pass.
The MLP is the part a mixture-of-experts model splits up.

THE TRAINING LOOP



repeated for trillions of tokens, over months

Why that loop needs a building full of chips

Arithmetic to run it
~2 FLOPs per parameter, per token

Memory to serve it
~2 bytes per parameter

Arithmetic to train it
~6 FLOPs the backward pass costs 2x the forward

Memory to train it
~16 bytes AdamW keeps two extra numbers per weight

Layer counts are typical of published open-weight frontier models; no closed model publishes its depth. The FLOP ratio is the standard accounting used across the scaling-law literature — roughly two floating-point operations per parameter per token to run a model forward, and roughly six to train, because the backward pass costs about twice the forward one. The memory figures are for mixed-precision AdamW as commonly implemented: about two bytes per parameter to hold weights for serving, against roughly sixteen to hold weights, gradients, a full-precision master copy and Adam's two running statistics for training. Both are orders of magnitude for teaching, not figures from any one company's disclosure.

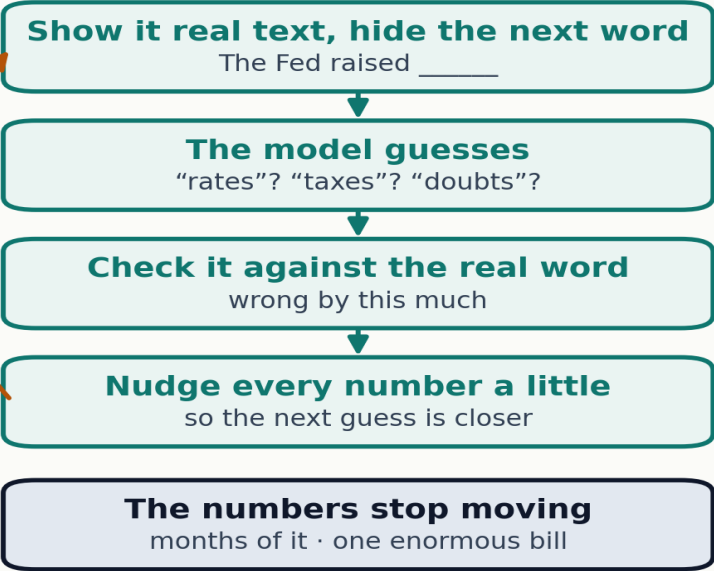
Training is writing the recipe book. Inference is cooking one meal from it.

A model is just a very large file of numbers. There are only two things you can do with it: set those numbers, or run them.

repeat trillions of times

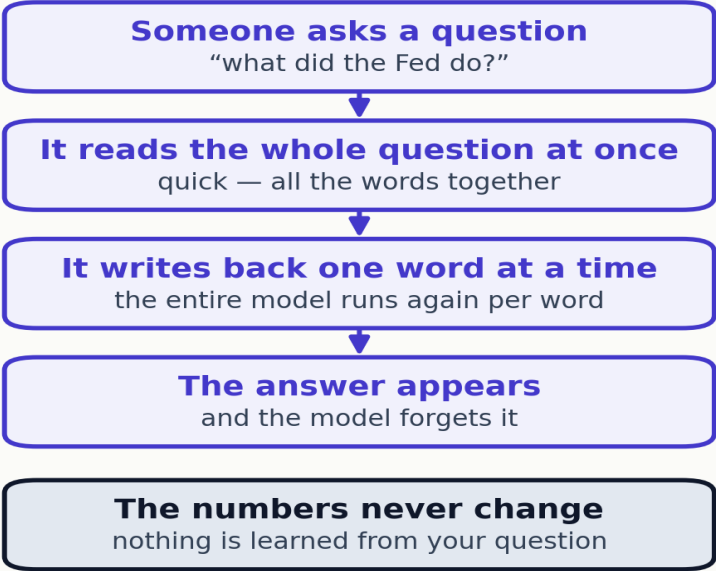
TRAINING

done once, before anyone can use it



INFERENCE

done again for every single question



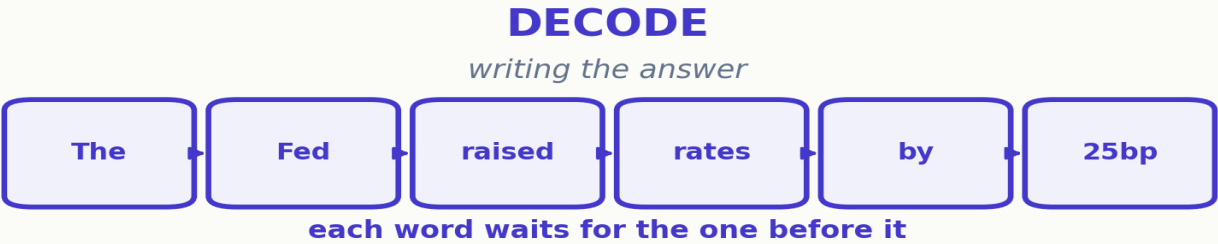
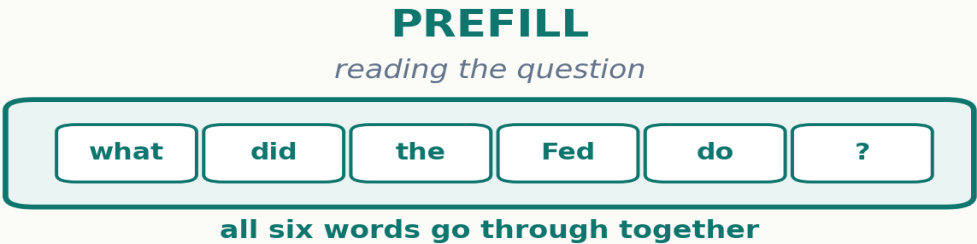
again per word

What it is	Making the model	Using the model
When you pay	Once, up front	Every time anyone asks
If use doubles	The cost is identical	The cost doubles

Simplified in two ways that matter later in this deck. Modern models route each token through only a fraction of their weights — the active parameters on Exhibit 1 — so a bigger model is not automatically a more expensive one to run. And a reasoning model writes a long hidden answer before the visible one, so a single question can cost many times what the reply on screen suggests.

Reading your question is one pass. Writing the answer is one pass per word.

Inference has two phases with completely different economics — and the price list shows it.



How many times the whole model has to be pushed through the chips



Busy doing arithmetic — compute-bound. Cheap.

Busy moving weights — bandwidth-bound. Not cheap.

Which is why every provider quotes two prices, and output is the dear one

	per million tokens in		per million tokens out	
Claude Sonnet 5	\$2	→	\$10	5x
Claude Opus 5	\$5	→	\$25	5x
Claude Fable 5.1	\$10	→	\$50	5x

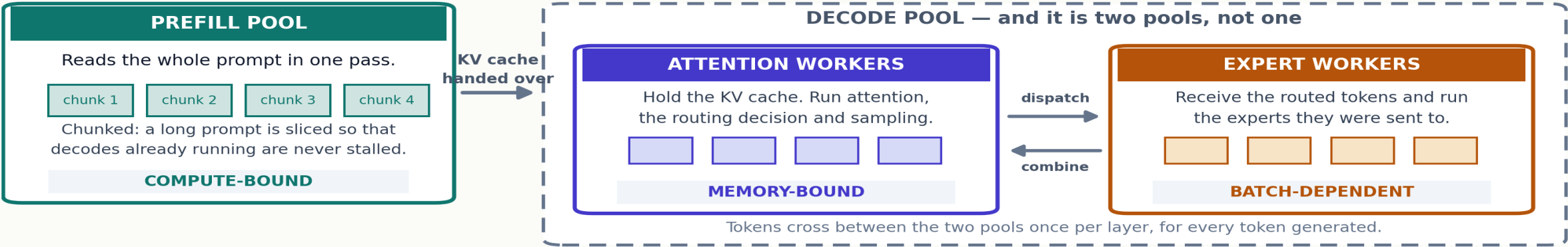
Writing is five times the price of reading, right across the range.

Prices are Anthropic's published list prices per million tokens, retrieved 20-Sep-2026; the same input-to-output ratio holds across the range. Other providers publish the same two-price structure. The model is not literally re-read from disk each time — the weights sit in high-bandwidth memory beside the chips — but they must be streamed through those chips once per output word, which is what makes decoding bandwidth-bound rather than compute-bound, and what makes a long answer expensive while a long question is cheap.

The two phases are now run on different machines — and decode is split again.

Prefill and decode want opposite things from a GPU, so they run on different machines now. Inside decode, attention and experts split too.

ONE REQUEST, THREE DIFFERENT POOLS OF MACHINES



The published work does not agree on what limits the expert side
MegaScale-Infer says sparse activation shifts MoE feed-forward networks from compute-intensive to memory-intensive: each expert sees too few tokens to fill a GPU.
A design-space study of the same technique instead calls the MoE stage compute-bound dense matrix multiplies, and puts the memory bound on attention. This exhibit does not pick one.

WHAT EACH SYSTEM REPORTS FOR ITSELF

2.6–5.6×
serving capacity, from chunked prefill
Sarathi-Serve, OSDI 2024, vs vLLM

1.90×
per-GPU throughput, attention split from FFN
MegaScale-Infer, SIGCOMM 2025

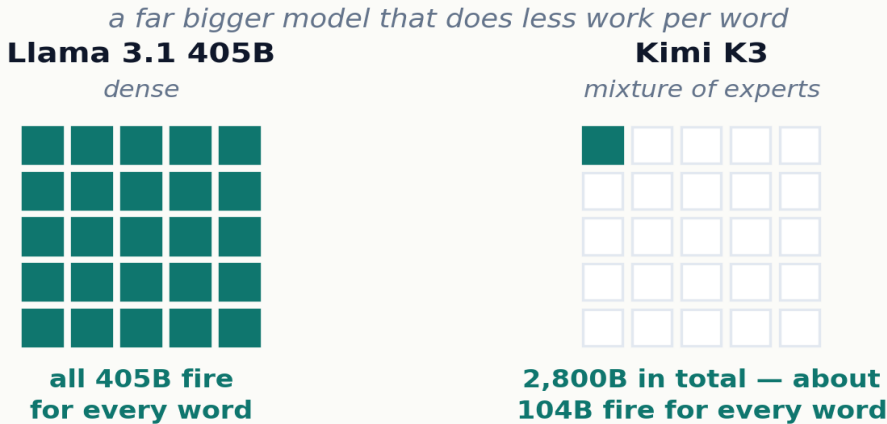
1.35–1.45×
per-GPU decode throughput, same split
FastAFD, on a GB200 NVL72

Sources. Chunked prefill and stall-free batching are from Sarathi-Serve (Agrawal et al., USENIX OSDI 2024), which reports 2.6× higher serving capacity on Mistral-7B on one A100, up to 3.7× on Yi-34B on two, and up to 5.6× end-to-end on Falcon-180B, all against vLLM under tail-latency limits. Attention–FFN disaggregation is from MegaScale-Infer (arXiv 2504.02263, ACM SIGCOMM 2025), reporting up to 1.90× higher per-GPU throughput, and FastAFD (Hao AI Lab, UCSD), reporting 1.41–1.44× per-GPU decode throughput on Qwen3-235B and 1.35–1.45× on MiniMax-M2.5, on a GB200 NVL72. Each figure is the reported result of the system that published it, on its own models and hardware; none is reproduced here and they are not comparable with one another. The sources disagree on what bounds the expert side, as the amber note says, and this exhibit does not pick between them. The three pools are drawn as a topology, not a capacity split: no public disclosure gives the ratio of machines an operator actually runs in each pool.

One change made each word cheaper. The other made models write far more words.

Mixture of experts broke the link between how big a model is and what it costs to run. Reasoning put the cost back.

MIXTURE OF EXPERTS



Seven times the size. A quarter of the work per word.

REASONING MODELS

the same price per word, many more words



You never see the amber part. You are billed for all of it.

Anthropic's smallest thinking budget is 1,024 tokens. Budgets run past 32,000 — all billed at the output rate.

So which one wins?

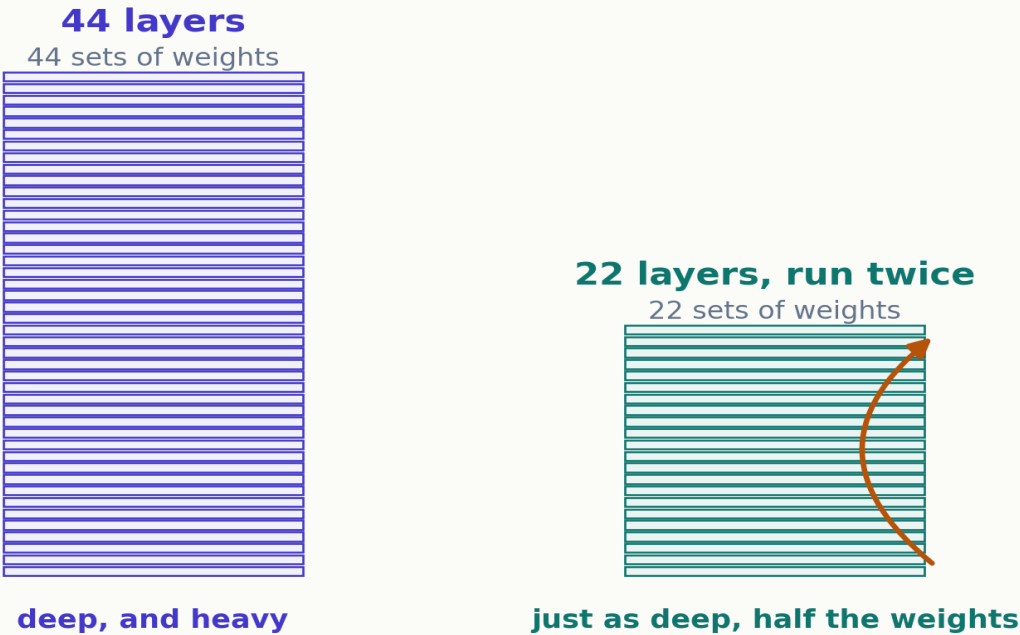
Price per token keeps falling. Tokens per answer keeps rising. At \$50 per million output tokens, a 32,000-token think costs \$1.60 before the model has written a single word you will actually read.

That is why “inference is getting cheap” is only half the story.

Parameter counts are the developers' own published figures, the same ones plotted on Exhibit 1: Llama 3.1 405B is dense, so all 405B parameters are used for every token; Kimi K3 is 2,800B in total with about 104B active per token. Thinking-budget figures and the billing treatment are from Anthropic's published documentation, retrieved 20-Sep-2026: the minimum budget is 1,024 tokens, budgets above 32,000 are directed to batch processing, and thinking tokens are reported as part of billed output tokens. The \$1.60 is arithmetic on those published numbers — 32,000 tokens at the \$50 per million output rate — not a figure any vendor publishes. Other labs describe the same structure in their own terms.

Depth without parameters: run the same layers twice

The same stack, reused. Twice the arithmetic, none of the extra memory — which is the right trade when memory is the bottleneck.



The trade

Depth of computation	the same
Memory footprint	half
Arithmetic per token	about double

Why that trade is attractive

Writing an answer is limited by moving weights, not by doing sums — so spend the sums.

GPT-6 ASTRA — REPORTED, NOT CONFIRMED

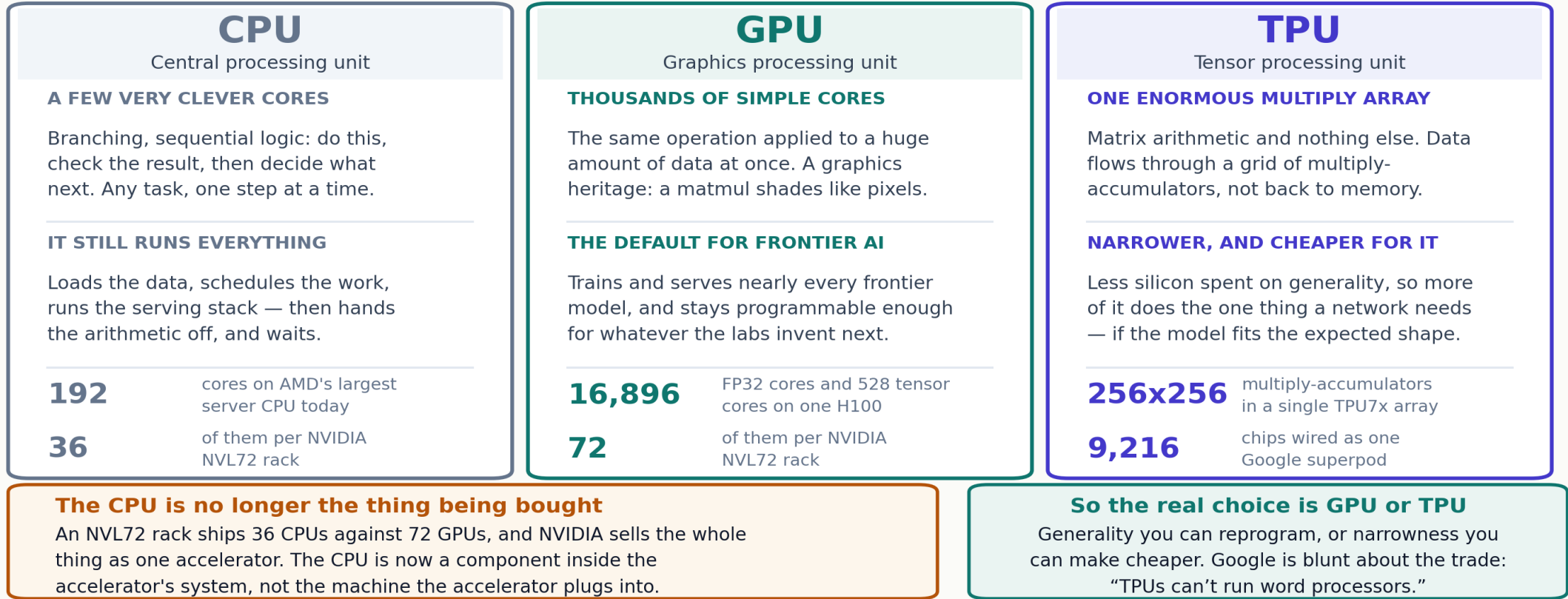
OpenAI has published no architecture for Astra. The looped-transformer claim comes from reporting. The example on the left is an open model that does publish its design.

Looping trades memory for arithmetic. Reasoning trades tokens for quality. Both are bets on which resource stays scarce.

Source: the 22-layer example is Nanbeige 4.2, an open-weight model whose architecture is published, as described by Sebastian Raschka. On GPT-6 Astra: OpenAI has published no architecture for it. The looped-transformer attribution comes from reporting, not from disclosure, and Raschka — who follows this closely — argues the looped aspect is being overstated: “Astra may be a really good model, but this shouldn't be about this ‘looped transformer aspect,’ which is just a tiny architectural tweak.” He also notes that reusing layers does not by itself hide a model's chain of thought. Treat any Astra architecture claim the way this deck treats every closed-model figure on Exhibit 1: reported, not confirmed.

Three kinds of processor sit in the same rack. The capital is going to two of them.

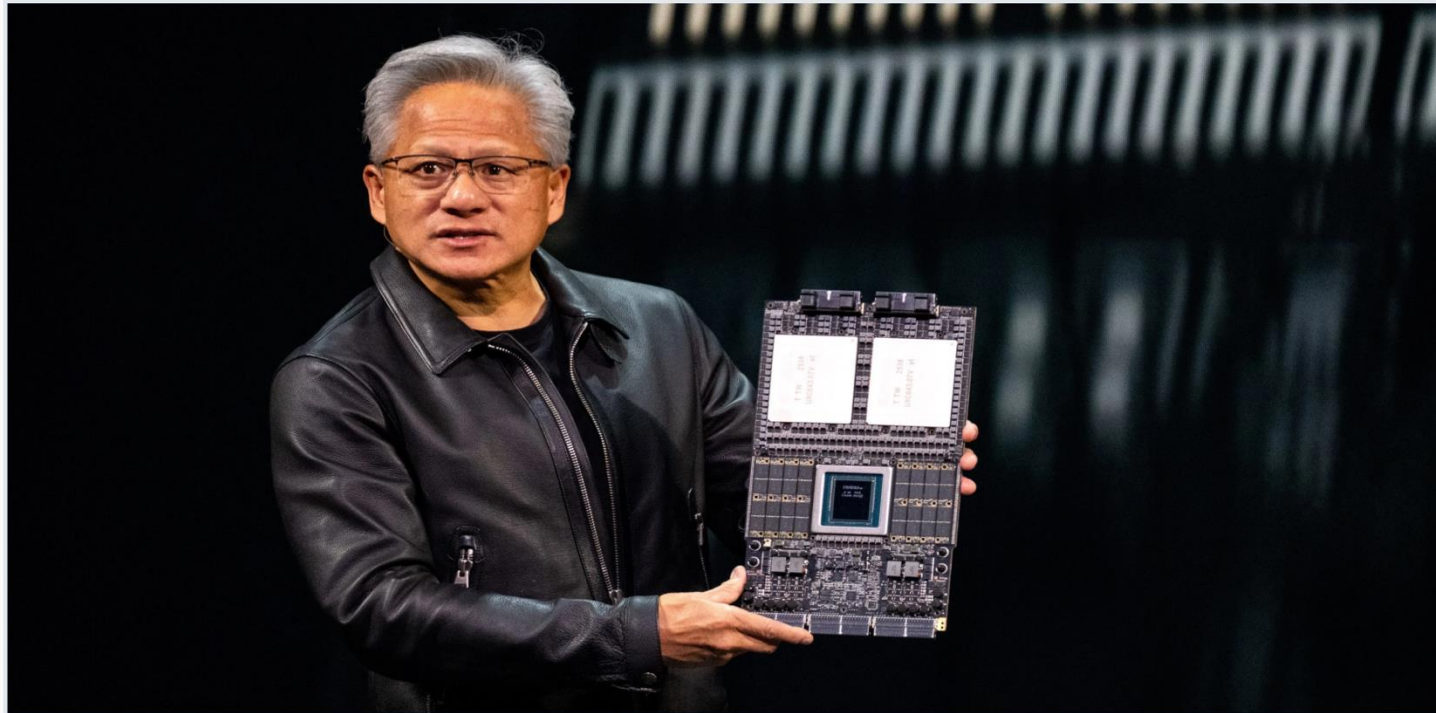
A CPU runs the program. A GPU and a TPU do the arithmetic inside it. The difference is how many things each can do at once, and how narrow the job is.



Sources: CPU core count from AMD's EPYC 9005 series product page — up to 192 'Zen 5' or 'Zen 5c' cores. GPU core counts from NVIDIA's H100 Tensor Core GPU architecture whitepaper: 132 SMs, 16,896 FP32 CUDA cores and 528 fourth-generation Tensor Cores per SXM5 GPU. Rack composition from NVIDIA's GB200 NVL72 product page: 36 Grace CPUs and 72 Blackwell GPUs, 2,592 Arm Neoverse V2 cores, 13.4TB of HBM3E at 576TB/s, 17TB of LPDDR5X at 14TB/s, 130TB/s of NVLink. TPU architecture and the quoted line from Google Cloud's TPU system-architecture documentation: an MXU is 256x256 multiply-accumulators in a systolic array on TPU v6e and TPU7x. H100 is cited because NVIDIA publishes per-SM core counts for Hopper and has not for Blackwell — it illustrates GPU structure, not the current generation. No performance comparison is made here.

Nobody buys a chip. They buy this, and then they buy seventy-two of them.

Jensen Huang holding an NVIDIA compute board on stage. The silicon is the small square in the middle.



WHY THE BOARD, NOT THE CHIP

A GPU on its own is not a product. NVIDIA sells the module — processors, memory and interconnect already assembled — and then the rack those modules slot into.

72

of these go into one NVL72 rack

All-to-all

NVLink, so the rack acts as one accelerator

260TB/s

of NVLink bandwidth across that rack

288GB

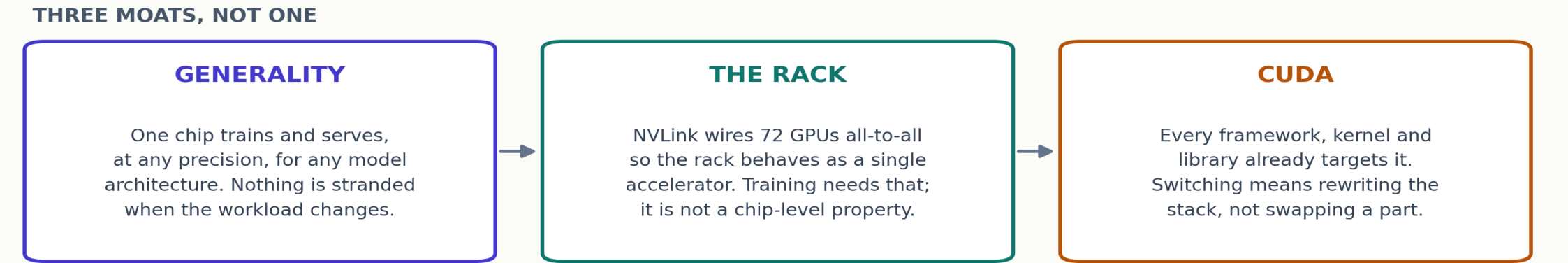
of HBM4 per GPU, at up to 22TB/s

The rack is the product. The chip is a component.

Photograph supplied for this deck, from an NVIDIA stage presentation. Confirm reuse rights with the rights-holder before publishing or circulating the deck outside a classroom. The specific product in the photograph is not identified here — the visible markings are test and serial labels, not a model name — so the figures on the right describe NVIDIA's current published generation rather than this exact board.

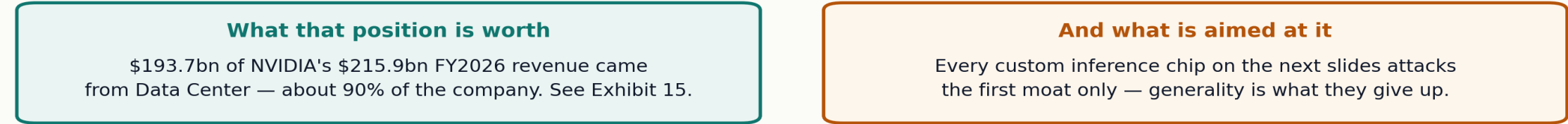
The product is not the chip. It is the rack, and the software that already targets it.

One accelerator that trains and serves anything, wired so that 72 of them behave as one — and a decade of code written for it.



The current generation, by NVIDIA's numbers — Vera Rubin

336bn transistors per GPU Blackwell had 208bn	288GB HBM4 per GPU at up to 22TB/s	3.6TB/s NVLink per GPU 260TB/s per rack	72 GPUs per rack wired as one accelerator
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Source: NVIDIA's own technical materials for the Vera Rubin platform. Specifications quoted are NVIDIA's: 336 billion transistors against Blackwell's 208 billion, up to 288GB of HBM4 per GPU at up to 22TB/s, NVLink 6 at 3.6TB/s bidirectional per GPU and 260TB/s across an NVL72 rack, and a Vera CPU of 88 custom cores linked to the GPUs at 1.8TB/s. The throughput figures — 50 PFLOPS of NVFP4 inference, 35 PFLOPS of training, and multiples of 5x inference and 3.5x training over Blackwell — are NVIDIA's own claims and have not been independently benchmarked here. Revenue is NVIDIA's FY2026 result, as on Exhibit 15. CUDA's position is an observation about the software ecosystem rather than a figure either NVIDIA or anyone else publishes.

Google did not build a faster chip. It built a machine the size of a building.

Seven generations of co-design. The interesting part is not the silicon — it is a network that routes itself around its own failures.

1

EVERY CHIP TOUCHES SIX OTHERS

the edges wrap around

A cube is $4 \times 4 \times 4 = 64$ chips, each wired to six neighbours — and the far face wired back to the near one.

2

THE POD IS THE UNIT OF COMPUTE

1 chip
64 chips = a cube
256 chips = a pod
9,216 chips = a superpod

One job runs across the whole superpod, which it addresses as a single machine.

3

AND THE FABRIC HEALS ITSELF

The links are light and the switch is reconfigurable. A failed cube is bypassed and a spare dropped in, without stopping the job.

THE CURRENT GENERATION, BY GOOGLE'S NUMBERS — IRONWOOD, TPU7x

192GB
HBM3E per chip
at about 7.4TB/s

1.2TB/s
interconnect per chip
to six neighbours

9,216
chips per superpod
144 cubes, one fabric

42.5
FP8 exaflops per pod
1.77PB of memory

Co-design, all the way down
Google designs the chip, the XLA compiler, the JAX framework, the Pathways runtime — and the models that run on all of it.

And it is now sold to a rival lab
Anthropic contracted for up to one million TPUs and well over a gigawatt, worth tens of billions, with capacity arriving in 2026.

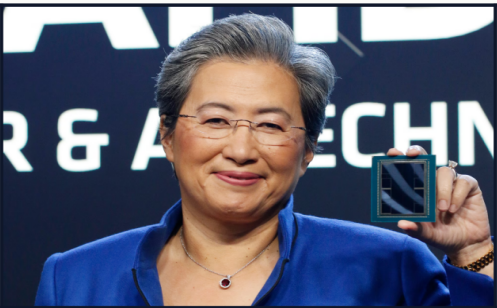
Source: Google Cloud's TPU7x documentation and its engineering post on the Ironwood stack. Per-chip figures are Google's: 192GB of HBM3E at about 7.4TB/s, 1,200GB/s of bidirectional inter-chip interconnect, and 2,307 BF16 or 4,614 FP8 teraflops. A pod is 256 chips in four cubes; a superpod is 9,216 chips in 144 cubes, holding 1.77PB of directly addressable memory and 42.5 FP8 exaflops — so a cube is 64 chips, which is the division of those disclosed totals rather than a separately stated figure. Google states Ironwood is about 30 times more power-efficient than its first Cloud TPU of 2018 and twice the performance per watt of the previous generation; both are Google's own comparisons and are not independently benchmarked here. Anthropic's commitment of up to one million TPUs and well over a gigawatt of capacity, worth tens of billions of dollars, was announced by Google Cloud on 23-Oct-2025 with capacity arriving in 2026. The lattice and the reroute are drawn schematically to show topology, not physical layout.

AI & Capital Markets

14

An agent takes six steps. One of them is the model. The other five run on a CPU.

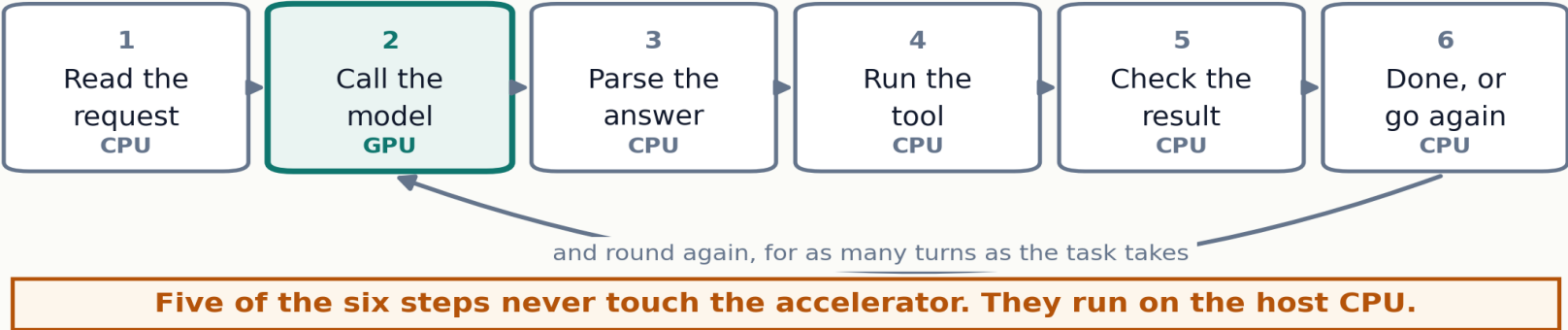
That is the part of the AI build-out AMD sells into, and it is why a server CPU is measured in cores and memory rather than teraflops.



AMD's chief executive, holding a processor package on stage.

No model marking is legible, so the part is not identified here.

ONE PASS OF AN AGENT



WHICH IS WHY THE HOST IS SOLD ON THESE NUMBERS

192 cores, and 384 threads, on one AMD EPYC part	6TB of DDR5 across twelve channels, in two sockets	160 PCIe Gen 5 lanes for the drives and the network
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Mercury Research puts AMD at 46.2% of x86 server CPU revenue in the first quarter of 2026, on 33.2% of units.

Sources: part figures are from AMD's own EPYC 9005 series processor datasheet — up to 192 cores and 384 threads, up to 12 DDR5-6400 memory channels, 6TB of memory in a two-socket server, and 128 PCIe Gen 5 lanes rising to 160 in two sockets. Market share is Mercury Research data for the first quarter of 2026 as reported by Tom's Hardware: 46.2% of x86 server CPU revenue on 33.2% of units. The six-step loop is a simplification of how agent systems are built, not a measured breakdown: neither AMD nor its customers report revenue or utilisation by workload, so no claim is made here about how the time or the money splits between the two kinds of chip. Photograph used on the basis that reproduction rights have been confirmed. The package in the photograph is not identified here — no model marking is legible in it.

The customer is now the designer

Sam Altman and Broadcom's Hock Tan with the first Jalapeño wafer, June 2026.



WHAT THEY ARE HOLDING

A wafer of Jalapeño dies — OpenAI's first custom chip, built with Broadcom and designed for one job: running its models.

~840 mm²

per die — near the 858 mm² reticle limit

9 months

from start to tape-out

Late 2026

first deployment, at gigawatt scale

Six

HBM modules alongside each die

Broadcom builds it. NVIDIA is not in the frame.

Photograph supplied for this deck; it accompanied the OpenAI-Broadcom announcement of 24-Jun-2026. Confirm reuse rights with the rights-holder before publishing or circulating the deck outside a classroom. The plaque in the photograph reads "Jalapeño Intelligence Processor". Die size and tape-out timing are from Tom's Hardware; the deployment timing is OpenAI's own. No benchmarks have been published.

Both labs are building silicon aimed squarely at the decode bottleneck

A training chip has to be good at everything. An inference chip only has to be good at writing one word at a time.

OPENAI · JALAPEÑO

ANNOUNCED · 24 JUN 2026 *with Broadcom*

“A blank-slate design for modern LLM inference, not a general-purpose accelerator adapted from earlier AI workloads.”

— OpenAI

- ~840 mm² compute chiplet, near the reticle limit
- Six HBM modules — it keeps the costly memory
- Nine months from start to tape-out
- Deploying from late 2026, at gigawatt scale

Claimed, not yet demonstrated
“performance per watt substantially better than current state-of-the-art” — no benchmarks published

ANTHROPIC

CONFIRMED · AUG 2026 *no chip yet*

Confirmed by Anthropic

- Developing a custom chip, co-designed with its own future models
- Building an in-house chip team

Reported, not confirmed

- Samsung as manufacturing partner
- An inference-first focus
- Any timeline at all

Why a chip, and why now

THE PROBLEM

Writing an answer is limited by moving weights, not doing sums. A GPU spends decode waiting.

THE FIX

Strip out everything a training chip needs. Spend the die on memory and data movement.

THE PRIZE

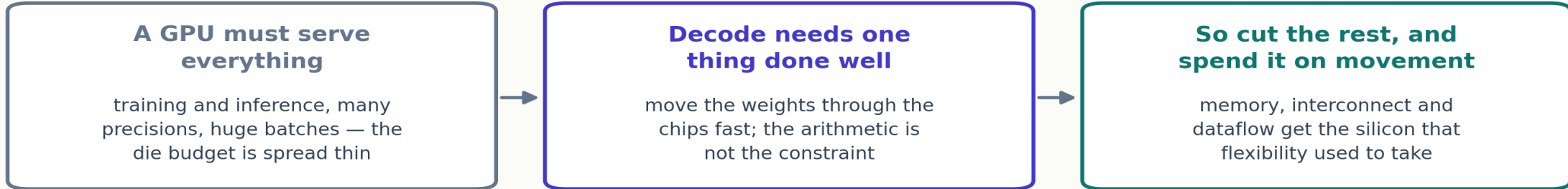
OpenAI's stated aim: utilisation “much closer to theoretical peak” — and a lower cost per token.

Sources: OpenAI's own announcement with Broadcom, 24-Jun-2026, for the design rationale, the deployment timeline and the performance-per-watt claim; Tom's Hardware for the die size, the six HBM modules and the nine-month tape-out. OpenAI has published no benchmarks — it says a detailed technical report will follow “in the coming months” — so the performance claim is the company's own and is unverified. Anthropic: a spokesperson confirmed to Business Insider in August 2026 that it is developing a custom chip co-designed with its future models and is building an in-house team. Samsung as manufacturing partner, an inference focus and any timeline are reported by other outlets and not confirmed by Anthropic; they are shown here as reported, not as fact.

Give up everything a training chip needs, and spend the silicon on moving weights

The design principle is public. Almost every number that would let you check it is not.

THE LOGIC — why a fixed-function chip can beat a GPU at this one job



What OpenAI and Broadcom have actually said

- “Reduce unnecessary data movement” and better match compute, memory and networking resources
- “Trades breadth for efficiency on a narrow set of tasks”
- Aims for the throughput of today's best accelerators with latency closer to specialised inference systems
- Not sold as a chip: OpenAI, Broadcom and Celestica are building finished racks, boards and networking
- Designed in nine months, with help from OpenAI's own earlier models; prototypes already run real workloads

THE CLAIM — UNVERIFIED

“Roughly 50% lower cost and roughly 50% better cost efficiency versus standard AI GPUs”
— Broadcom's chief executive. No benchmarks published.

WHAT HAS NOT BEEN DISCLOSED

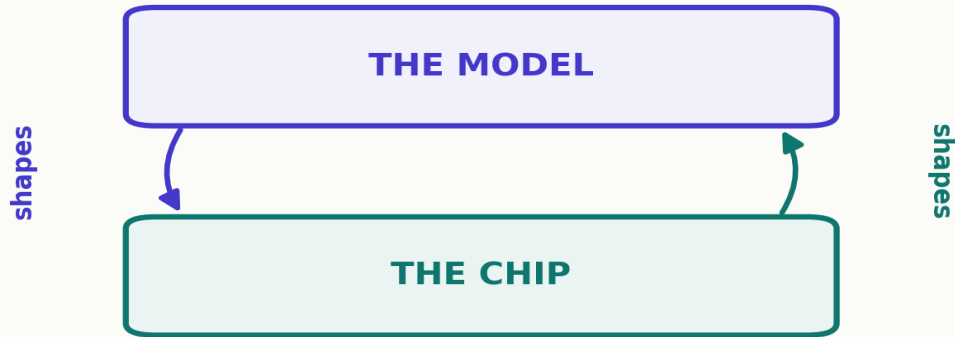
- | | |
|-------------------------------|---------------------|
| × fabrication node | × transistor count |
| × HBM generation | × memory bandwidth |
| × how it handles the KV cache | × batching strategy |
| × number formats | × pricing |

Sources: OpenAI's announcement with Broadcom, 24-Jun-2026, for the design language quoted; RCR Wireless and VentureBeat for the rack-level integration with Celestica, the nine-month tape-out, the use of OpenAI's own earlier models in the design work, and the prototype workload. The cost claim is Broadcom's chief executive, reported by RCR Wireless and separately by Bloomberg; neither OpenAI nor Broadcom has published benchmarks, and OpenAI says a technical report will follow. The three steps across the top are the general engineering logic of a fixed-function inference accelerator, not a description of Jalapeño's internals, which have not been disclosed. Treat the whole slide as a design intent that has not yet been independently measured.

Own the model and the chip, and each one can be shaped around the other

A merchant chip has to guess the workload years ahead. A lab designing its own already knows what it intends to ship.

THE LOOP



Normally only the top arrow exists: hardware is fixed and the model bends to fit it. Owning both means the chip can bend too.

And the loop already closed once

OpenAI used its own earlier models to help design the chip — nine months to tape-out.

WHAT IT BUYS YOU

- ▶ **You already know the workload**
The chip is “architected around OpenAI's vision for the future of LLM inference” — not a guess at what the market will want in three years.
- ▶ **The model can move to suit the chip**
Anthropic is explicit about this: it will “co-design the processor with its future large language models.”
- ▶ **Nothing is spent on what you never use**
No die area for training, for precisions you don't run, for workloads you'll never serve.
- ▶ **The whole system ships as one**
It isn't sold as a chip. OpenAI, Broadcom and Celestica are building the racks, boards and networking together.

THE PRIZE

Performance per watt and per dollar. Broadcom's chief executive claims roughly 50% lower cost — unverified.

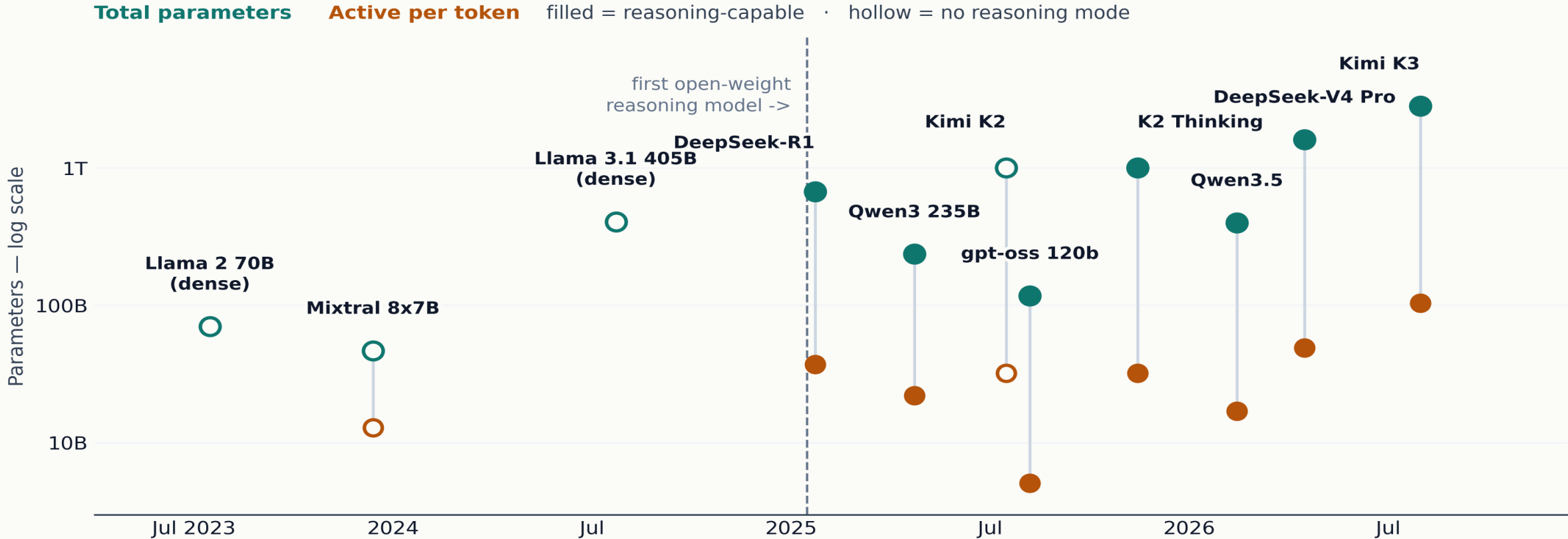
THE RISK

You carry the design cost, with no merchant volume to spread it over — and a fixed-function chip ages badly if model architecture moves. See the useful-life slide.

Sources: OpenAI's announcement with Broadcom, 24-Jun-2026, for the design language and the rack-level integration; RCR Wireless and VentureBeat for Celestica's role, the nine-month tape-out, the use of OpenAI's own earlier models in the design work and the GPT-5.3-Codex-Spark prototype workload; a spokesperson quoted by Business Insider, August 2026, for Anthropic's co-design intent and its use of AI in verification. The cost claim is Broadcom's chief executive and is unverified — no benchmarks have been published by either company. The costs listed at the foot are the standard economics of custom silicon rather than statements by either company, and the depreciation point connects to the useful-life slide earlier in this deck.

Exhibit 1: Models grew ~40x in total size — but compute per token barely moved

Mixture-of-experts split size from cost: a 2.8T model now fires only ~4% of its weights per token.



Source: developers' own model cards, technical reports and release posts (Hugging Face, Meta AI, Mistral, DeepSeek arXiv 2412.19437, OpenAI, Moonshot AI, Alibaba/Qwen). Only officially disclosed counts are plotted. Closed frontier models — GPT-5.x, Claude, Gemini, Grok 4.x — are absent because no lab has disclosed their parameter counts; figures circulating for them are leaks or estimates. Reasoning status is the developer's own description; hybrid models with a thinking toggle count as reasoning-capable.

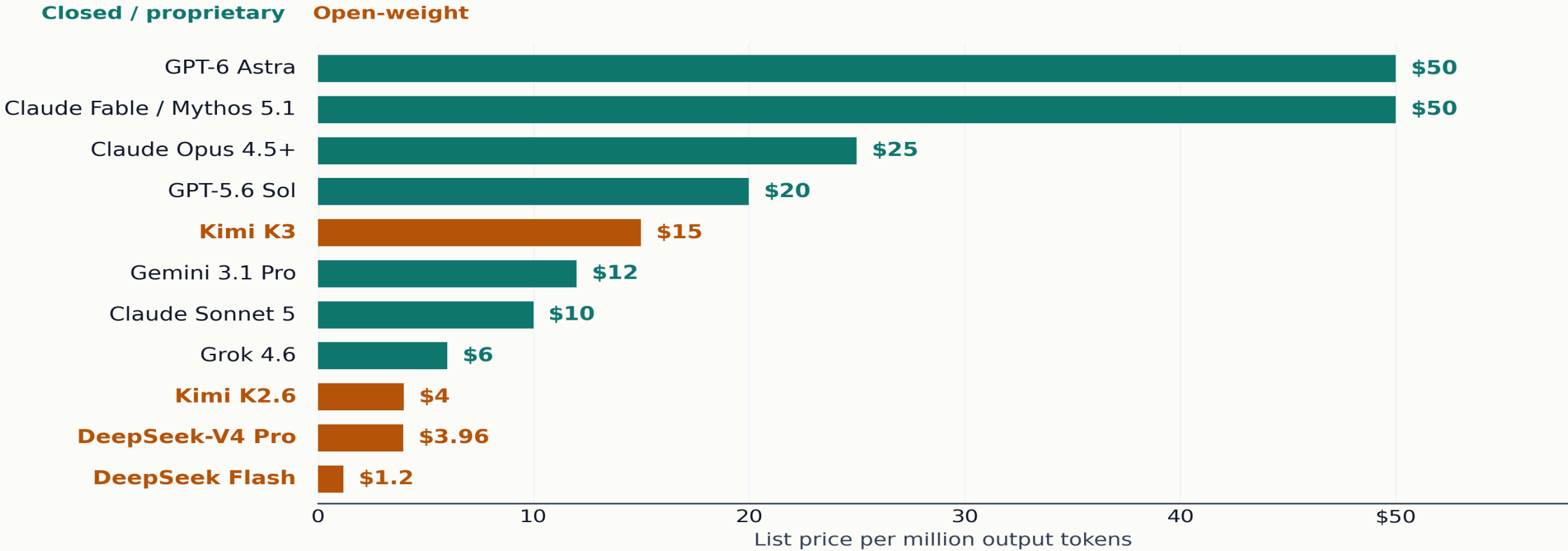
Training is a capex event; inference is a running cost

- Training: one huge compute bill, months long, paid up front.
- Inference: pennies per query, but billions of queries.
- GPUs, data centers, and power serve both — hence the buildout.

Finance lens: training $\approx$ building the plant; inference $\approx$ operating it.

Exhibit 2: Open-weight models undercut the closed frontier many times over

Closed flagships list at \$25–\$50 per million output tokens. Open-weight models run \$1.20–\$15.



Source: each provider's own published pricing page, read 20 Sep 2026 — OpenAI, Anthropic, Google Vertex AI, xAI, DeepSeek, Moonshot AI. Output tokens only; input tokens cost less. List prices exclude caching and batch discounts. DeepSeek figures are peak-hour rates (off-peak is half). Gemini 3.1 Pro is the ≤200K-context tier. Open-weight prices are one provider's hosted API — those models can also be self-hosted, which is part of why the tier prices where it does.

02

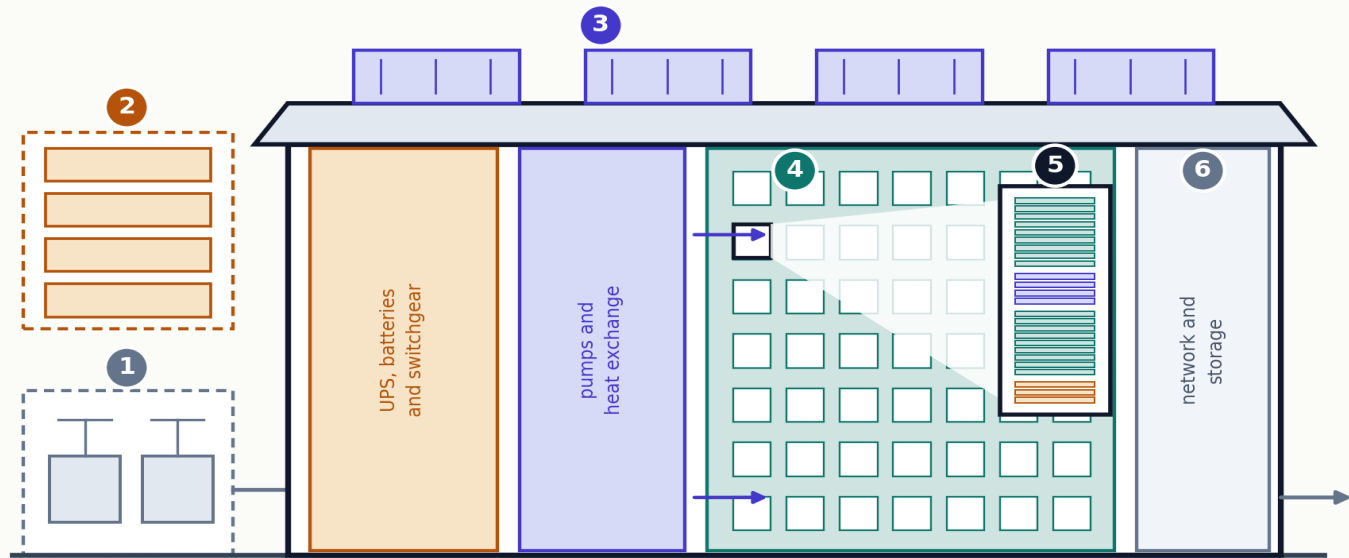
Why costs exploded

Scaling laws showed that more compute reliably buys better models — so spending became the strategy.

Cut a data centre open and most of it is not computers.

Six parts. Only the fourth and fifth do any arithmetic; the rest exist to power them, cool them and connect them.

ONE BUILDING, CUT OPEN



PART BY PART

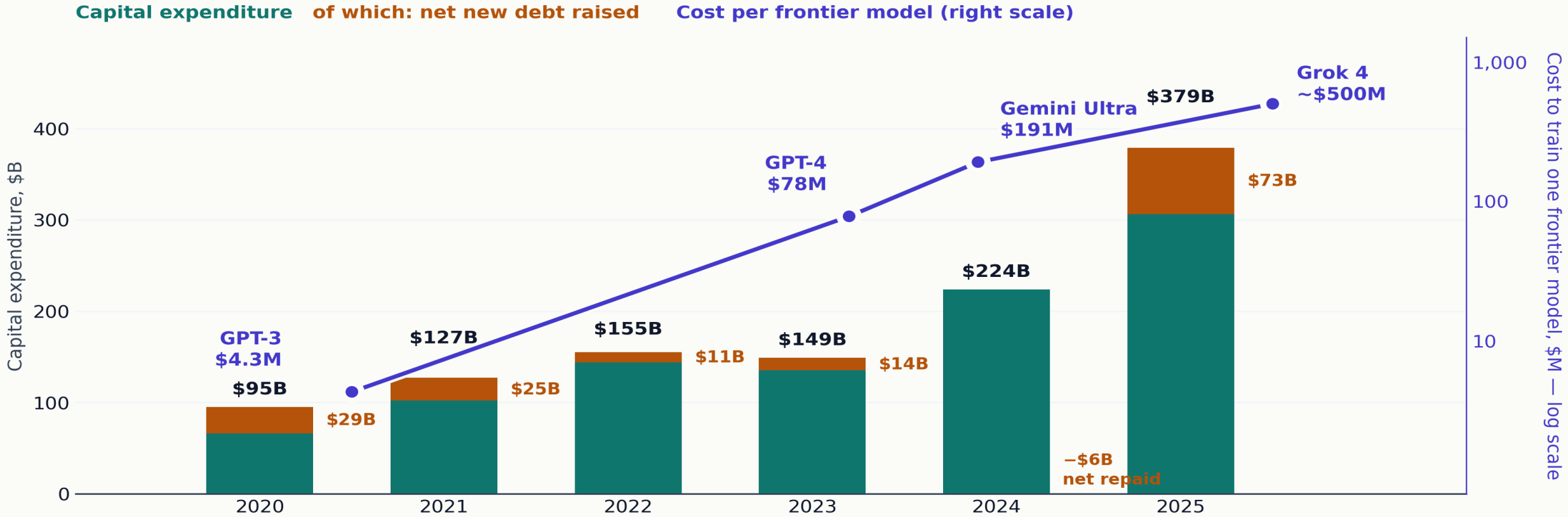
- 1 GRID CONNECTION**
A substation in the hundreds of megawatts. The longest lead time on the whole project.
- 2 POWER THAT NEVER DROPS**
Generators outside, batteries and switchgear inside, so the racks never notice the grid flicker.
- 3 COOLING**
Chillers on the roof, pumps and heat exchange in a bay. At AI density the water goes to the chip.
- 4 THE DATA HALL**
The white space: rows of racks, cold aisle in and hot aisle out. The reason the building exists.
- 5 ONE RACK**
18 compute trays, 9 NVLink switch trays, 8 power shelves of 33kW. Up to 142kW, liquid cooled.
- 6 NETWORK**
Fibre out to the other buildings, so several campuses can train a single model together.

1.54 average PUE across the industry against 1.09 for Google's fleet	142kW for one AI rack over 80% of sites top out at 30kW	176TWh used by US data centres in 2023 a projected 325-580TWh by 2028
---	--	--

Sources: rack figures from NVIDIA's NVL72 AI Factory enterprise reference architecture — a GB300 NVL72 rack holds 18 compute trays carrying 72 GPUs, 9 NVLink switch trays and 8 power shelves of 33kW; it is liquid cooled and requires up to 142kW. Efficiency and density figures from the Uptime Institute's 2025 Global Data Center Survey: a weighted average annual PUE of 1.54, flat for a sixth year; 46% of operators report racks averaging 1-7kW and over 80% of facilities no rack above 30kW. Google's 2025 fleet-wide PUE of 1.09 is its own disclosure. National figures are from the Department of Energy's release of Lawrence Berkeley National Laboratory's 2024 US Data Center Energy Usage Report: 176TWh and 4.4% of US electricity in 2023, and a projected 325-580TWh by 2028 — a projection, not an outcome. The cutaway is schematic: room layouts, proportions and equipment counts differ by operator, and nothing here is drawn to scale or from a published floor plan.

Exhibit 3: Debt receded from the buildout, then came back in 2025

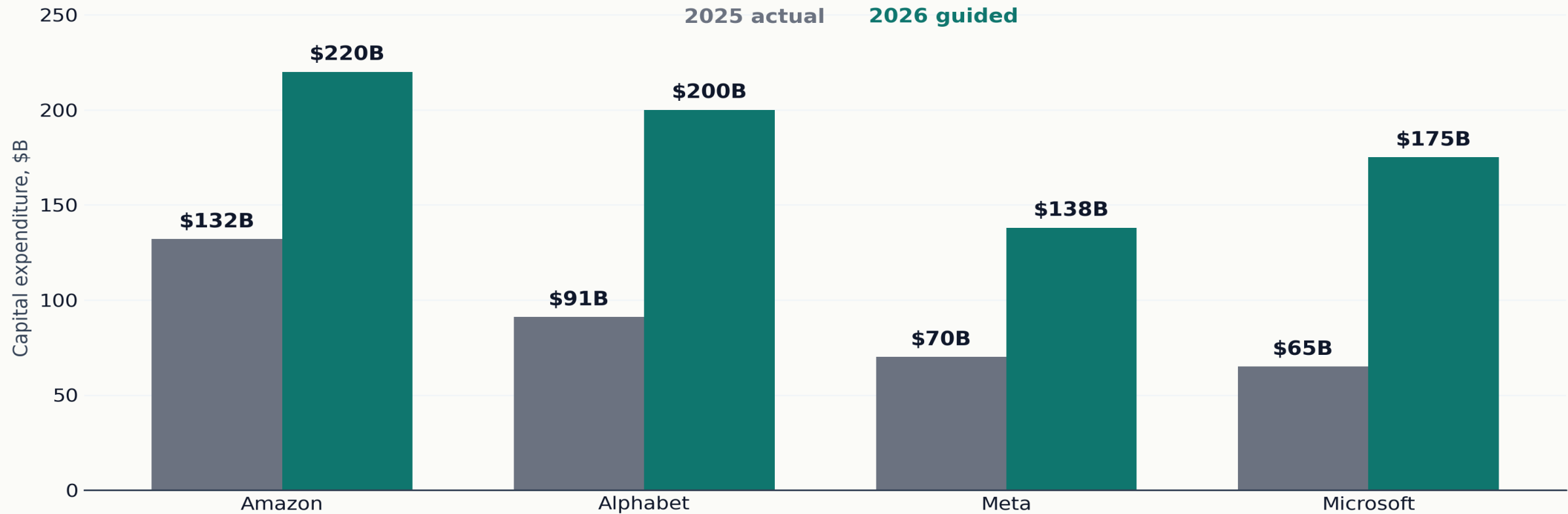
Net new borrowing swung from –\$6B in 2024 to \$73B in 2025 — 19% of capex, close to its 2021 share.



Source: SEC 10-K consolidated statements of cash flows (EDGAR / XBRL) for Microsoft, Alphabet, Amazon, Meta and Oracle. Capex = purchases of property and equipment, excluding finance leases (Amazon gross). Net new debt = proceeds from debt issuance less repayments — net, not gross, because Alphabet's line includes commercial paper; in 2024 the five repaid \$6B more than they raised. Microsoft (FY to Jun) and Oracle (FY to May) are fiscal years. Training costs: Stanford AI Index 2024 / Epoch AI.

Exhibit 4: Big-four capex roughly doubles from 2025 to 2026 guidance

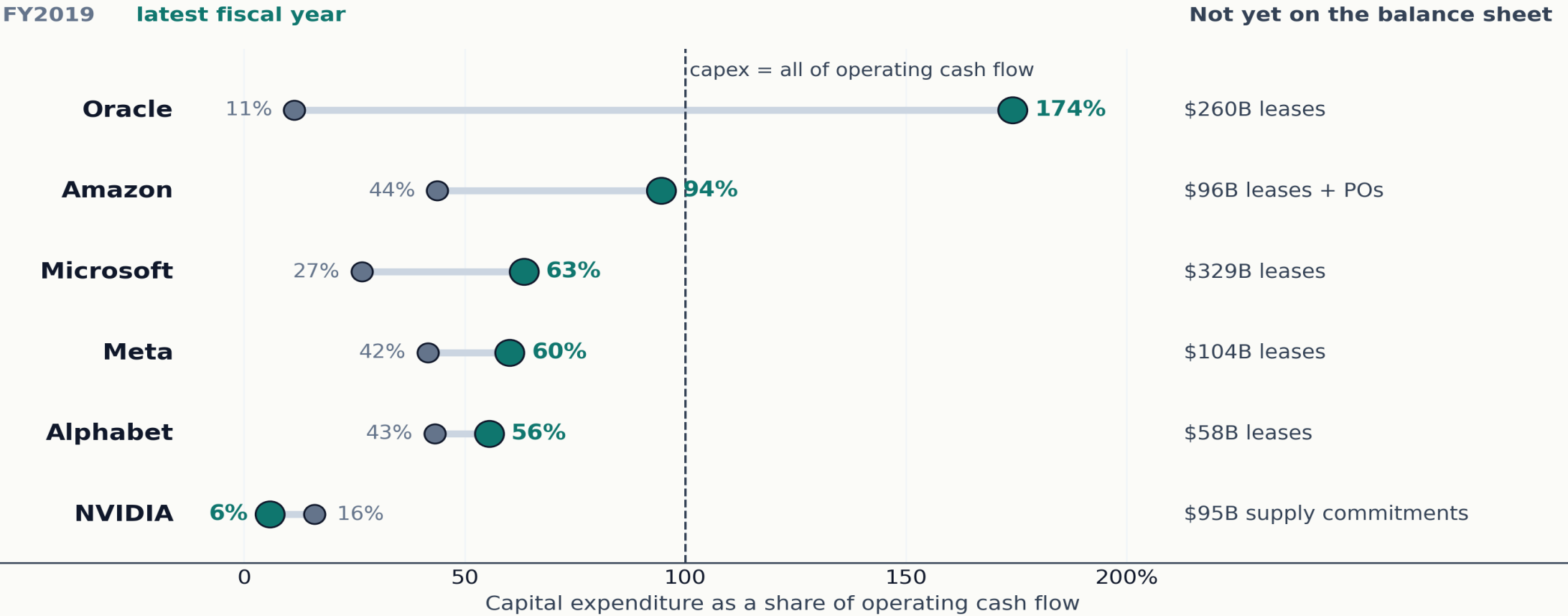
Big four only — Exhibit 3 adds Oracle: \$358B in 2025 → \$720–745B guided for 2026.



Source: 2025 actuals from SEC 10-K cash flow statements — purchases of property and equipment, excl. finance leases; Microsoft is FY to June 2025. 2026 = midpoint of company guidance (Aug 2026).

Exhibit 5: Capex now eats most of big tech's operating cash flow

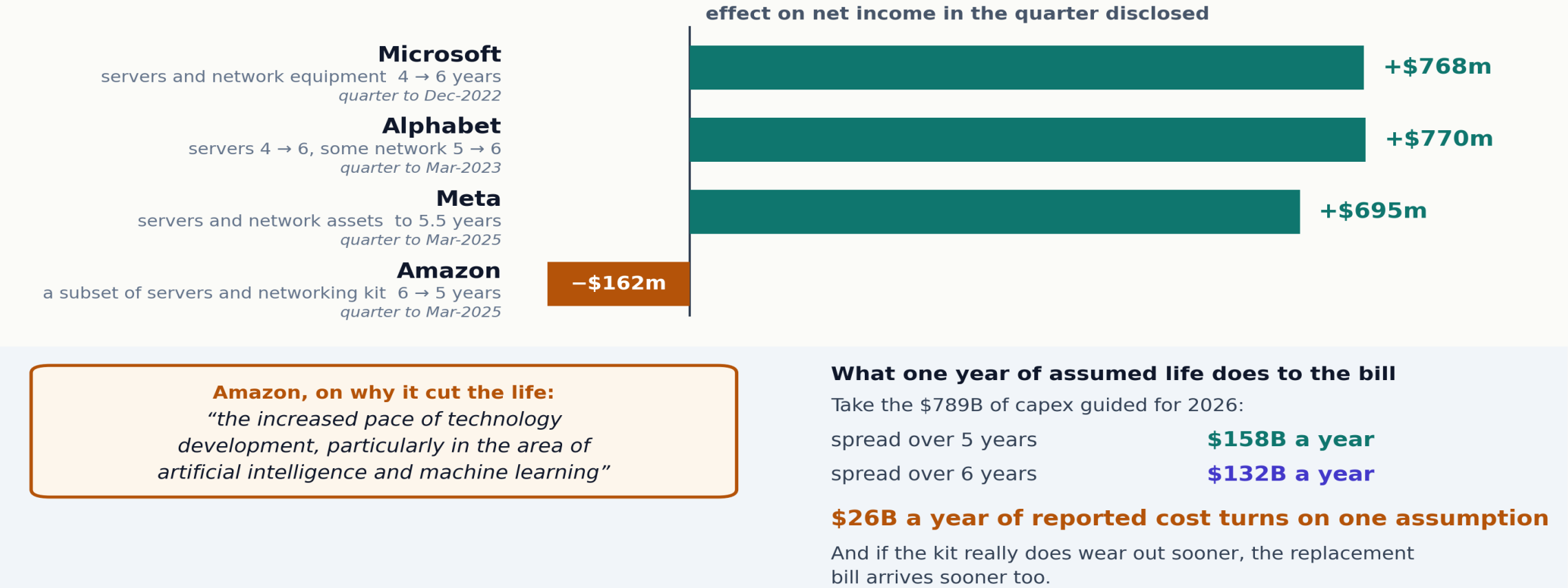
Oracle spends 174% of it. NVIDIA, which sells the hardware rather than buying it, spends 6%.



Source: SEC 10-K cash flow statements (EDGAR XBRL). Capex = purchases of property and equipment; revenue and net cash from operating activities as filed. Ratios are derived. Latest fiscal year: Microsoft FY2026 (Jun-26), Oracle FY2026 (May-26), NVIDIA FY2026 (Jan-26), Alphabet / Amazon / Meta FY2025 (Dec-25) — period-ends span nine months. Capex EXCLUDES leases, which is how most data-centre capacity is now financed — the right-hand column shows commitments disclosed but not yet on the balance sheet. NVIDIA's capex line includes intangibles.

One accounting assumption, and billions of reported profit either way

Three hyperscalers extended the life they assume for their servers, which lifted reported earnings. Amazon shortened its, and blamed AI.



Source: the filing in which each company disclosed the change. Microsoft Form 10-Q for the quarter ended 31-Dec-2022 (change effective FY2023); Alphabet Q1 2023 results exhibit (effective FY2023); Meta Form 10-Q for the quarter ended 31-Mar-2025 (effective 1-Jan-2025); Amazon Form 10-Q for the quarter ended 31-Mar-2025 (effective 1-Jan-2025). Each bar is the net income effect that company disclosed for that single quarter — the quarters are two years apart, so the bars are comparable in kind, not in period, and none is a full-year figure. Meta's disclosure states the new 5.5-year life without restating the prior one, so no starting point is shown for it. The depreciation arithmetic at the foot is this chart's own, applied to the \$789B of 2026 capex on the story chart: a straight-line charge over five years against six, before salvage value. Real schedules are not uniform across asset classes, so treat it as an order of magnitude, not a forecast.

\$3T

Morgan Stanley's estimate of data center and hardware spend through 2028
— only about half coverable from big-tech cash flow.

Morgan Stanley, July 2025. The other half is why debt markets enter the story.

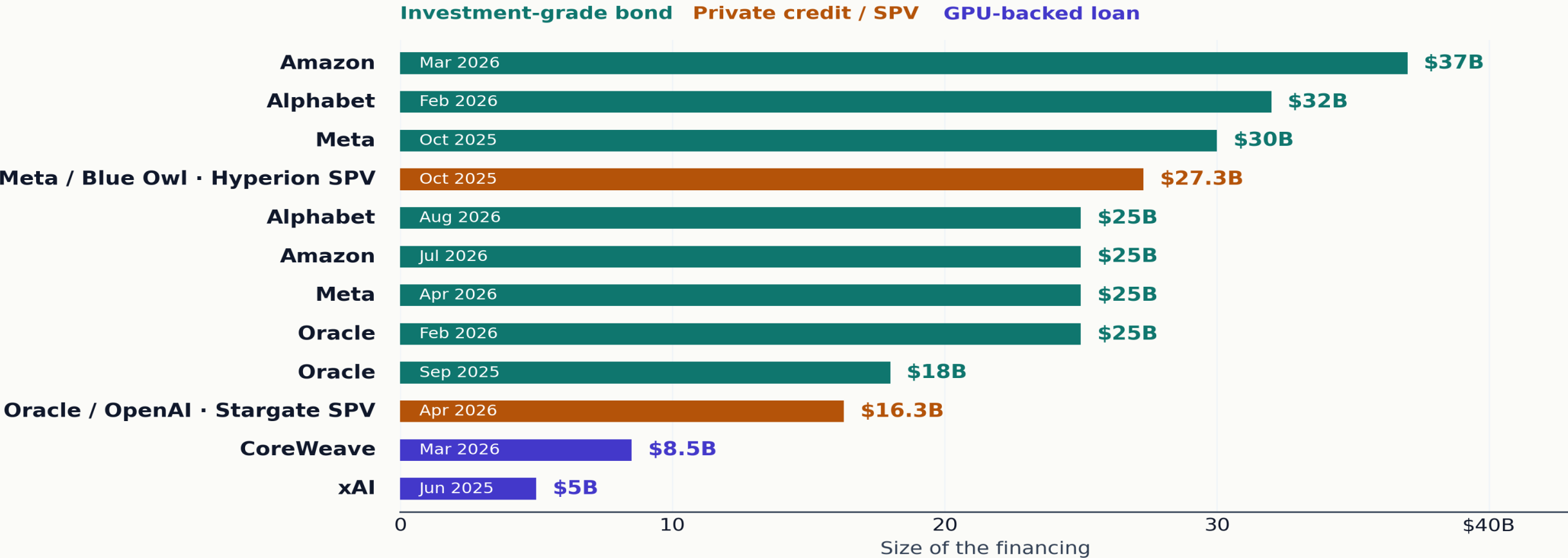
03

Debt & equity step in

When internal cash flow stops covering the bill, the buildout becomes a capital markets event.

Exhibit 6: AI infrastructure now drives the largest deals in credit markets

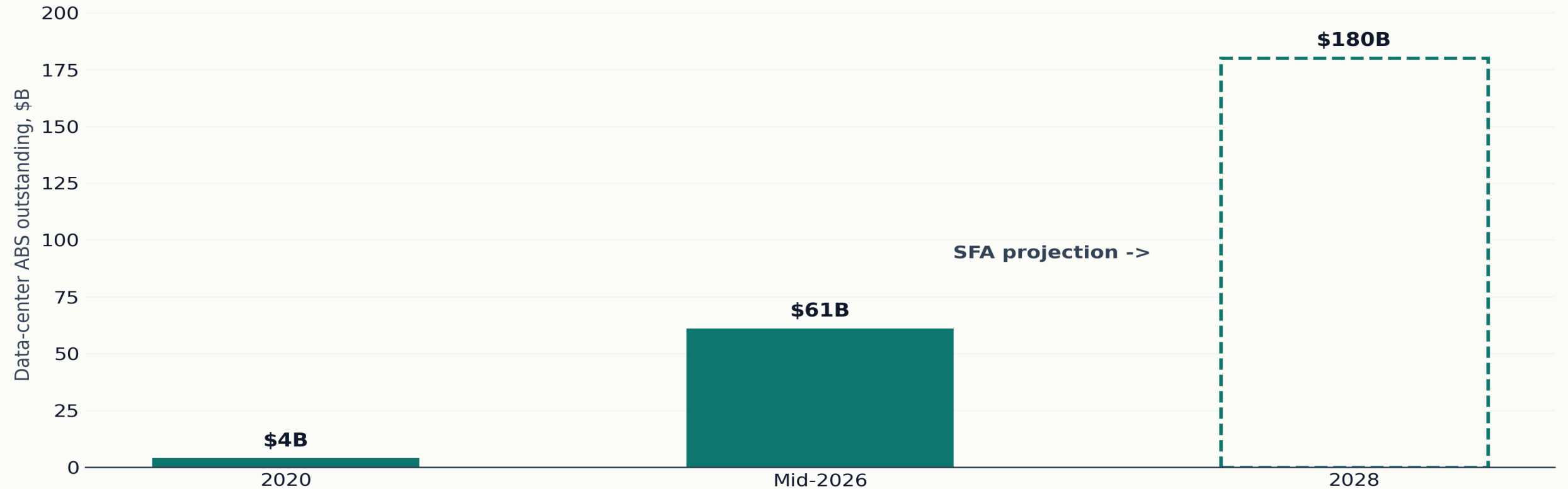
Eight of the twelve biggest are plain investment-grade bonds. The rest are private credit and GPU-backed loans.



Source: issuer press releases and SEC filings where available (Meta, Oracle, CoreWeave, Beignet Investor LLC 8-K), otherwise Bloomberg, CNBC and IFR reporting. Single transactions only — balance-sheet totals such as CoreWeave's cumulative debt are excluded. Amazon's Mar-2026 figure is the USD tranche; a euro tranche priced alongside it and its final size was not confirmed. Alphabet's Feb-2026 figure combines USD, sterling and Swiss franc tranches. Meta / Blue Owl is the \$27.294B senior secured note issue, not the wider development cost.

Exhibit 7: Data centers became an asset class — ABS up 15x since 2020

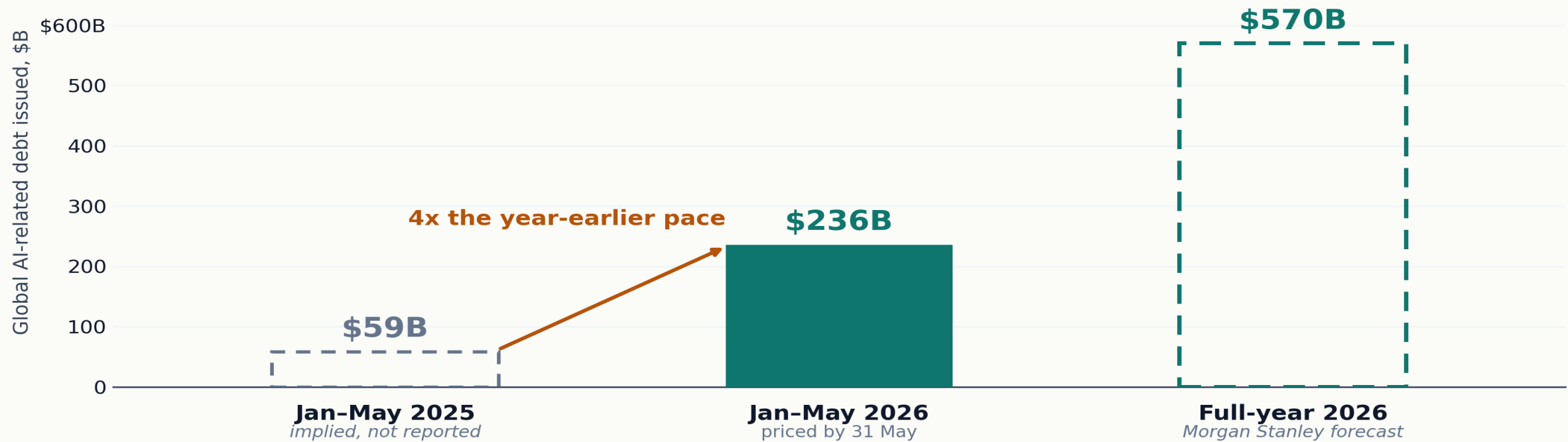
Private credit sits alongside: PIMCO alone took ~\$18B of the notes in Meta's \$27B Hyperion SPV.



Source: Structured Finance Association, July 2026. 2028 bar is an SFA projection, not outstandings.

Exhibit 8: AI borrowing is on track to be a \$570B year

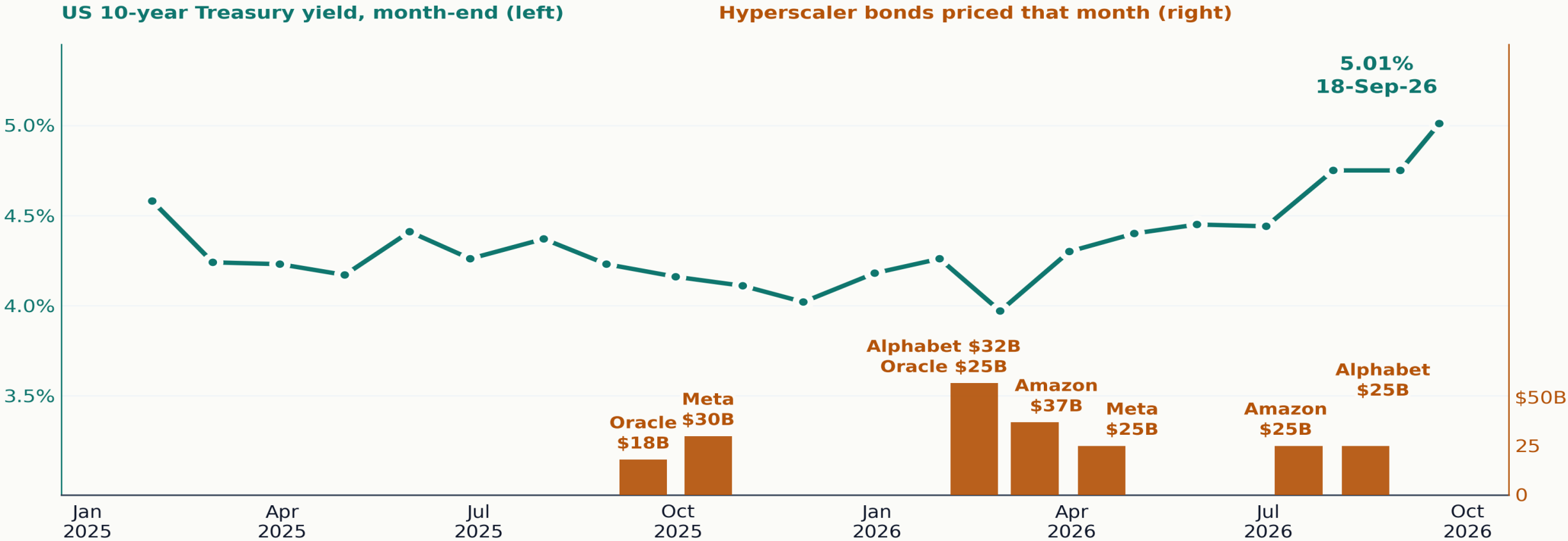
Global AI-related debt issuance ran four times the year-earlier pace through May. The buildout is now a debt-market event.



Source: Morgan Stanley research, reported by Reuters, 10-Jun-2026 — "nearly \$236 billion" of AI-related debt priced globally by 31-May-2026, "fourfold more than the same period last year", with full-year 2026 issuance on track for "nearly \$570 billion", more than twice the 2025 total. AI-related is Morgan Stanley's own classification, spanning investment-grade and high-yield bonds, private credit, securitisations and bank loans across hyperscalers, neoclouds, chipmakers and data-centre developers; the firm does not publish the split between them, so none is shown here. The Jan-May 2025 bar is hollow because it is not a published figure — it is what the fourfold comparison implies. Of which, for scale: the five hyperscalers sold \$121B of dollar bonds in 2025 (Morgan Stanley) and \$219B of investment-grade bonds in 2026 to date, \$62B of it in other currencies (J.P. Morgan Asset Management). Those are a subset of the totals above, not an additional amount.

Exhibit 9: Every hyperscaler bond now competes with a 5% Treasury

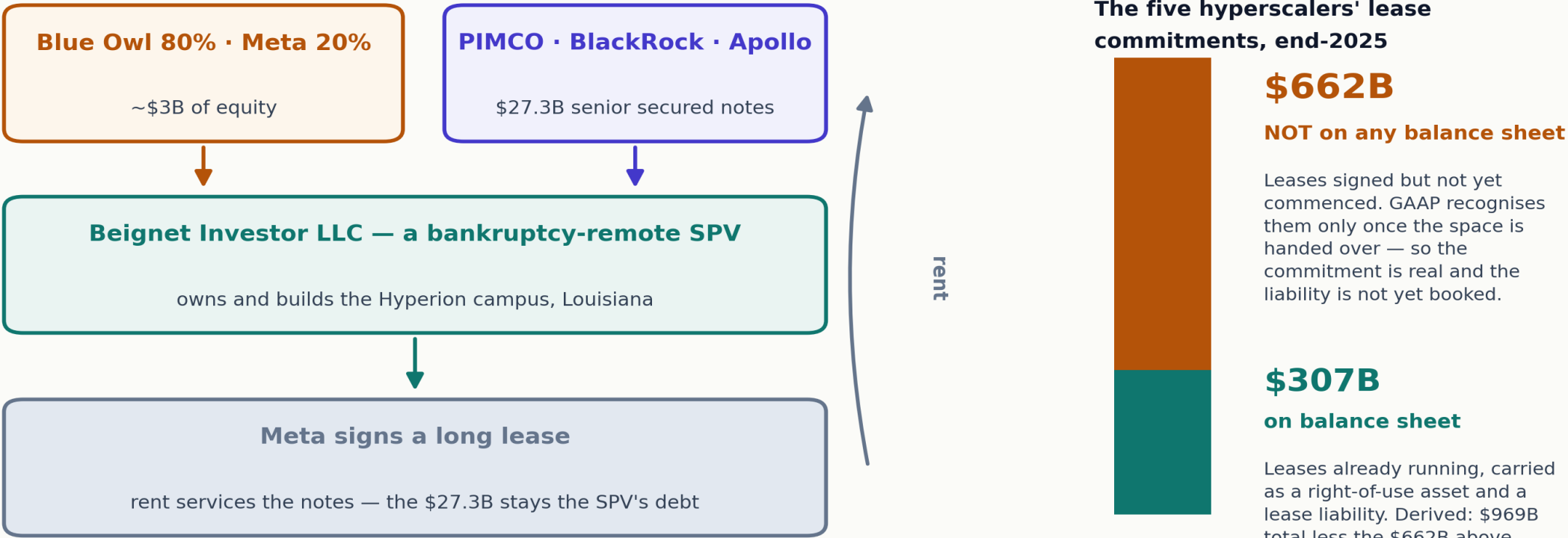
Eight deals since September 2025 raised \$217B. Over the same stretch the 10-year went from 4.16% to 5.01%.



Source, yield: US Treasury, Daily Treasury Par Yield Curve Rates, 10-year column, last business day of each month; September 2026 is the 18th, the latest published. Deal sizes are the transactions ranked in Exhibit 6 — SEC filings where available (Oracle "consummated" \$18B of notes 26-Sep-25; Meta \$30B priced 30-Oct-25), otherwise Bloomberg and IFR. Bars are grouped by month of pricing because Alphabet and Oracle both priced in February 2026. Amazon's March figure is the \$37B dollar tranche; a €14.5B tranche priced the next day, taking that offering to about \$54B globally. Alphabet's February figure combines its dollar, sterling and Swiss franc tranches. Microsoft, the fifth hyperscaler, has done no public bond deal in this window and so appears nowhere here.

How it works: the hyperscaler signs a lease, not a loan

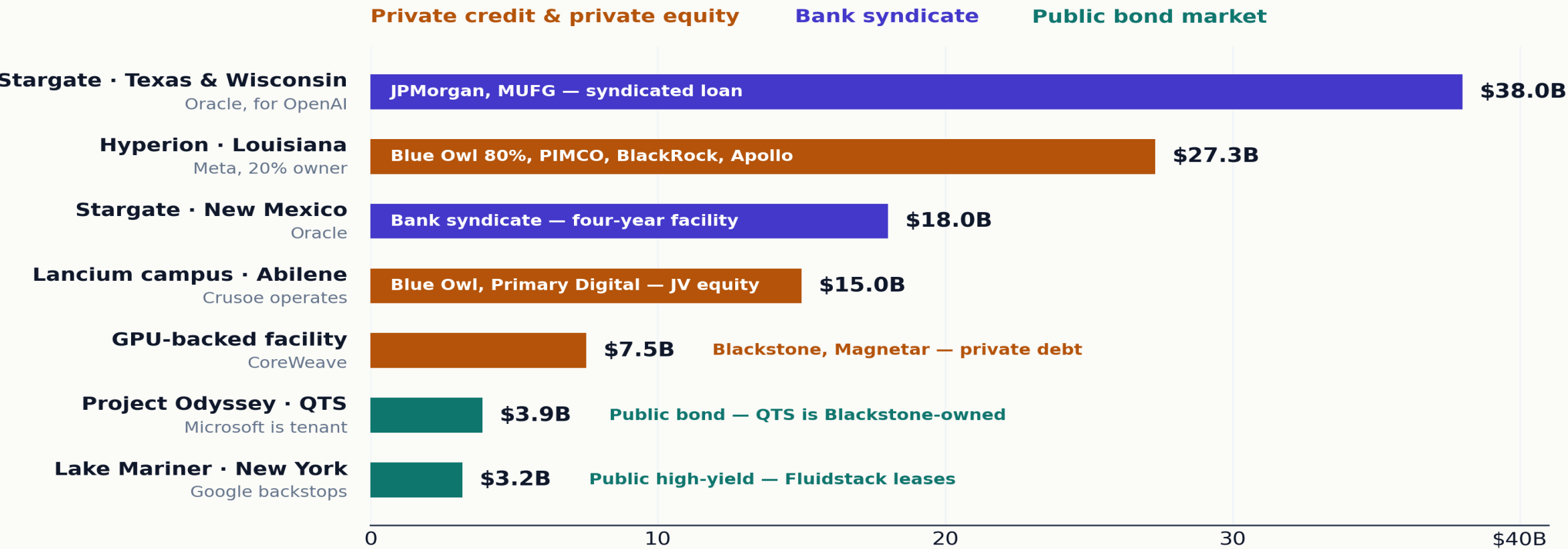
A separate company borrows and owns the site. The hyperscaler rents it. The debt is the borrower's; the rent is a commitment.



Source, structure: Beignet Investor LLC 8-K and the 144A offering for Meta's Hyperion campus, Richland Parish, Louisiana — \$27.294B of fully amortising senior secured notes due 2049, alongside roughly \$3B of equity split Blue Owl 80% / Meta 20%. Source, aggregate: Moody's Investors Service, February 2026, reported by Fortune and IFR — the five hyperscalers (Alphabet, Amazon, Meta, Microsoft, Oracle) disclosed \$969B of total undiscounted future lease commitments at end-2025, roughly double a year earlier, of which \$662B relates to leases that have not yet commenced and so is not recognised on balance sheet under GAAP. The \$307B on-balance-sheet figure is the difference between those two Moody's numbers, not a separately published one. Moody's has said it may make a quantitative debt adjustment for these structures; S&P accepted off-balance-sheet treatment for the Beignet structure. Alphabet alone disclosed \$42.6B of not-yet-commenced data-centre leases at Q3 2025, from \$23.9B a quarter earlier.

Exhibit 10: The largest AI sites are financed by everyone except the hyperscaler

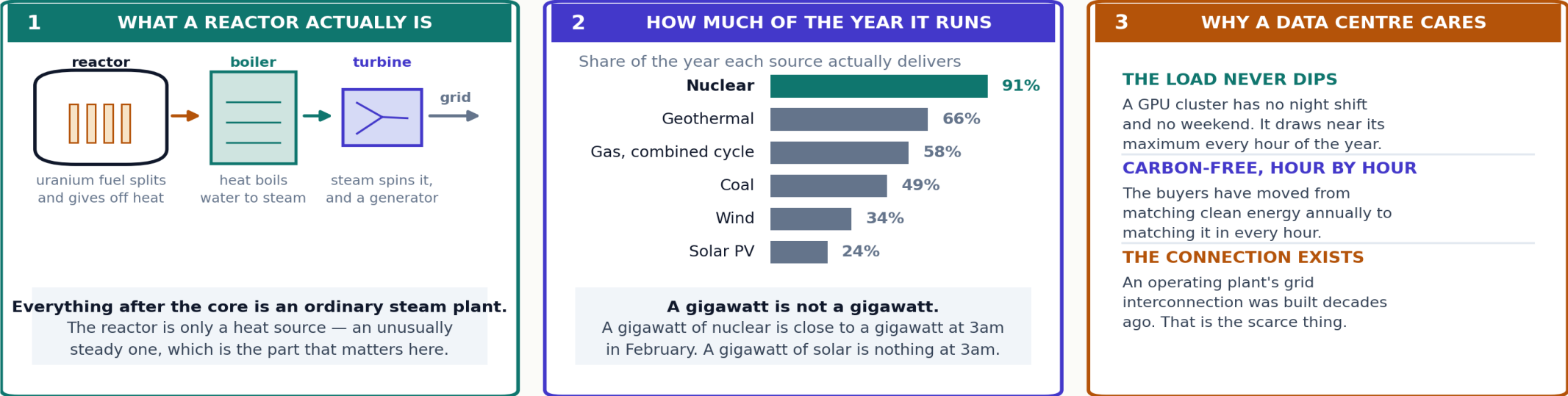
Blue Owl, Apollo, PIMCO, Blackstone and a bank syndicate hold the paper. The hyperscaler signs a lease.



Source: sponsor and arranger announcements — Crusoe/Blue Owl/Primary Digital JV press release (21-May-2025); CoreWeave and Blackstone press releases (May-2024); Beignet Investor LLC 8-K for Meta's Hyperion SPV; Bloomberg for the Oracle-tied bank packages (Oct-2025) and the QTS "Project Odyssey" notes (priced 18-Aug-2026, 7.63% five-year, ~\$10B of indications, Baa3/BBB-); Barron's and DCD for TeraWulf. Bars are capital committed to each structure, which is not a single instrument: the Abilene figure is the JV's total size and is mostly equity, Meta's is the \$27.294B senior secured note issue (a further ~\$3B of equity sits alongside it), and the rest are debt facilities or bond issues. They are therefore comparable in scale, not in seniority. Abilene's campus was subsequently leased by Oracle for OpenAI; that lease is not part of the JV announcement. Meta's Hyperion and the Oracle packages also appear in Exhibit 6, which ranks financings by size regardless of where the debt sits.

A reactor is a very steady kettle. That steadiness is the whole point.

A data centre draws almost its full load every hour of the year, so what matters is not a plant's nameplate but how much of the year it actually delivers.



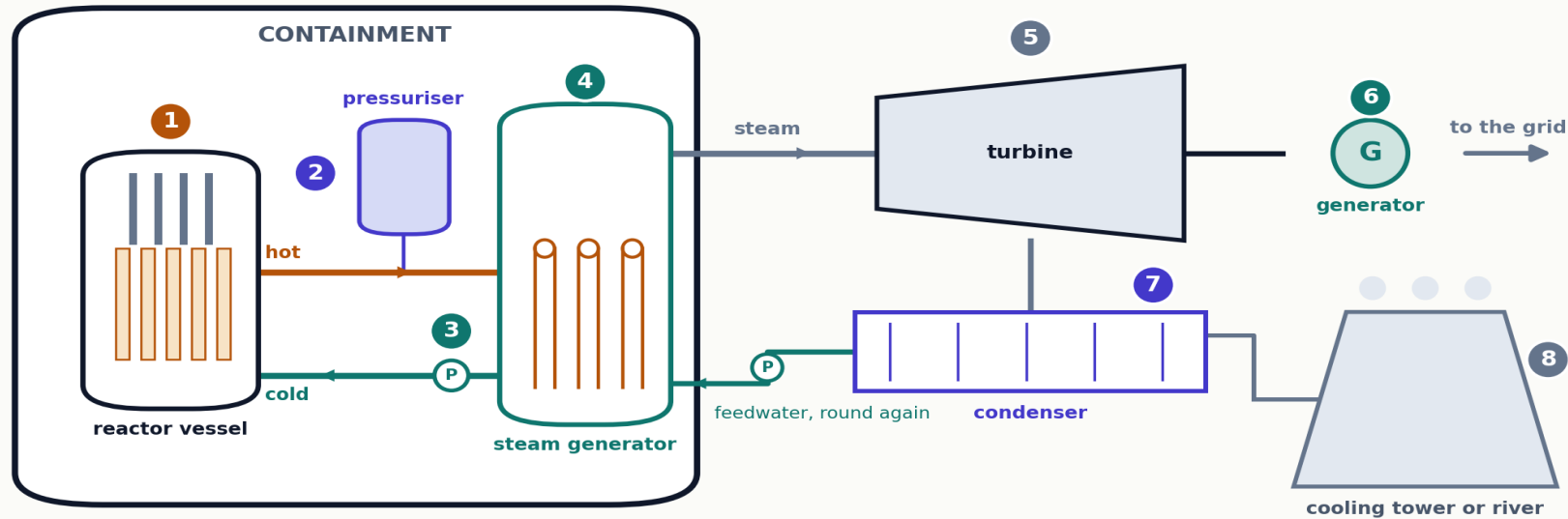
But most of it is not new electricity, and the new reactors are next decade

Contracting an operating plant's output to a data centre moves a megawatt rather than adding one — somebody else now buys from elsewhere. Only a restart genuinely adds supply, and only one of the deals on the next slide is a restart. The reactors that would add capacity outright are targeted for 2030 to 2039, while the buildings are being energised now, so gas and the existing grid carry the load in between. What nuclear is really securing is the 2030s — and, as the next slide shows, the contract signed today is what pays for it.

Sources: capacity factors are Energy Information Administration data for 2025 for utility-scale generators, as compiled by Visual Capitalist and retrieved 21-Sep-2026; the EIA's own Electric Power Monthly table was returning a server error at the time of writing, so the figures are shown on that secondary compilation rather than on the primary table. They are consistent with the Department of Energy's Office of Nuclear Energy, which states nuclear ran more than 92% of the hours in 2020 and puts it at roughly twice gas and coal and about three times wind and solar. Capacity factor is output over a period divided by the output the plant would have produced running flat out, so it measures availability, not efficiency or cost, and says nothing about what the electricity costs to produce. The reactor schematic is generic: it is a pressurised-water arrangement in outline and is not drawn to scale or from any particular plant's design. No levelised-cost comparison is made here, because it would turn on assumptions about capital cost, financing and lifetime that no source on this slide supplies.

Three water circuits, and the one that touches the fuel never leaves the building.

A pressurised water reactor, the commonest design. Heat on the left, electricity on the right, and a wall of tubes keeping the two apart.



Why three circuits and not one

The water that touches the fuel is radioactive, so it never leaves containment. It gives up its heat through the tube wall and turns straight back. The steam that reaches the turbine has never been near the core — which is why, in the NRC's words, most of the radioactivity stays in the reactor area.

PART BY PART

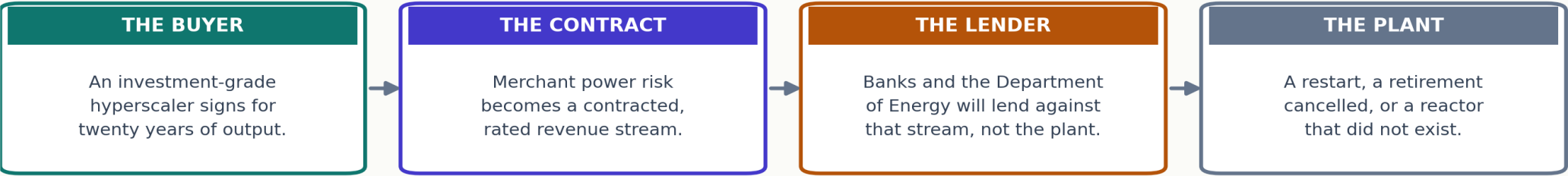
- 1 THE FUEL**
Uranium atoms split and give off heat. The grey rods absorb neutrons to slow it.
- 2 THE PRESSURISER**
Holds the primary water above 500°F but under pressure, so it never boils.
- 3 THE PUMP**
Round it goes: out at about 617°F, back at about 556°F, in again.
- 4 THE STEAM GENERATOR**
Primary water inside U-tubes, other water boiling outside them. They never mix.
- 5 THE TURBINE**
Steam spins high- and low-pressure turbines mounted on one shaft.
- 6 THE GENERATOR**
The shaft turns it; a transformer steps the voltage up for the grid.
- 7 THE CONDENSER**
Spent steam is cooled back to water and pumped round to be boiled again.
- 8 THE COOLING CIRCUIT**
A third, separate circuit carries the leftover heat away.

Source: the Nuclear Regulatory Commission's own material — its public description of pressurised water reactors and its training document ML11223A195. The NRC states that PWRs "keep water under pressure so that it heats, but does not boil", that "water from the reactor and the water in the steam generator that is turned into steam never mix", and that "in this way, most of the radioactivity stays in the reactor area". From the training document: the pressuriser "maintains the pressure of the reactor coolant at a high value, which prevents the high temperature (>500°F) coolant from boiling"; inside the steam generator "reactor coolant flows through U-tubes and transfers heat to the feedwater", and those tubes "act as a barrier between the primary and secondary cycles"; the reactor coolant pump "returns the reactor coolant to the reactor vessel, completing the primary cycle"; and the high and low pressure turbines are "mounted on a common shaft that drives the main generator". The 617°F and 556°F figures are that document's worked example for one four-loop unit at full power, not a universal specification. The diagram is schematic: one loop is drawn where a real plant has two to four, and nothing is to scale.

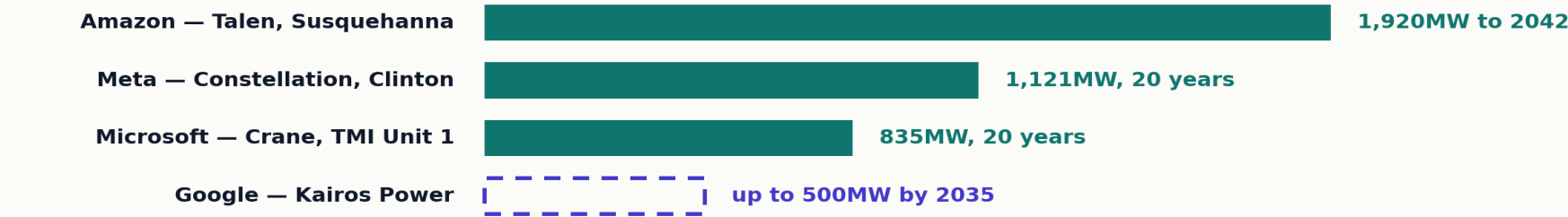
The hyperscalers are not buying power. They are lending their credit rating to it.

A twenty-year contract from an investment-grade buyer turns merchant power risk into bankable cash flow. That contract is what gets the plant financed.

WHY A POWER CONTRACT IS A FINANCING



WHAT THAT CREDIT HAS BOUGHT, IN MEGAWATTS



Solid: output from plants that already exist — one restarting, and one that was due to retire in 2027.
Dashed: reactors not yet built, so an announced target rather than contracted output.

~\$1.6bn
cost of the Crane restart
of which a \$1bn DOE loan, Nov 2025

~\$500m
Amazon into X-energy
Series C-1, October 2024

20 years
the standard contract length
on all three operating-plant deals

AND ONE AMBITION
5,000MW+
Amazon and X-energy, by 2039.
Not on the scale to the left:
none of it is built.

Sources, each the counterparty's own release. Crane Clean Energy Center (Three Mile Island Unit 1): Constellation, 20-Sep-2024 — about 835MW, a 20-year PPA with Microsoft, expected online in 2028; Utility Dive reports the restart may come as soon as 2027, a project cost of about \$1.6bn, and a \$1bn Department of Energy loan closed on 19-Nov-2025. Clinton Clean Energy Center: Constellation, 3-Jun-2025 — 1,121MW plus about 30MW of uprates, a 20-year term with Meta beginning June 2027. Susquehanna: Talen Energy, 11-Jun-2025 — 1,920MW to Amazon Web Services running through 2042 with extension options; no contract value is disclosed in that release. Kairos Power: Google, 14-Oct-2024 — up to 500MW, first reactor by 2030 and the rest through 2035. X-energy: 16-Oct-2024 — Amazon anchored roughly \$500m of Series C-1 financing, targeting more than 5GW of Xe-100 capacity online by 2039. The two dashed entries are reactors that do not yet exist: they are announced targets, not contracted output, and are drawn apart from the operating fleet for that reason. No total contract values are plotted because only Talen's counterparty disclosed capacity without price, and none of the five published a contract value.

Nebius: sign the contract, then borrow against it

A company earning \$582m a quarter spent \$5.7bn on hardware in that same quarter. The gap is the capital market.

WHAT IT RAISED



convertible notes

Aug-2026 · upsized from \$4.5bn



convertible notes

Sep-2025 · with an equity offering



senior secured debt

Jul-2026 · SOFR + 2.50%, due 2030

What the secured debt is pledged against
“deployed GPU infrastructure and contracted cash flows from an agreement with an investment-grade customer”

WHAT BACKS IT

META · up to \$27bn

Five years, announced Mar-2026: \$12bn of dedicated capacity from early 2027, and up to \$15bn of further purchases.

MICROSOFT · size not disclosed

Announced Sep-2025. Nebius calls it a multi-year, multi-billion dollar agreement and no more.

\$40bn

of customer commitments, as Nebius described them when discussing future financing.

\$582m	\$5.7bn	\$3.0bn	\$8bn
Q2 2026 revenue	Q2 2026 capex	run-rate revenue	cash at quarter end
+454% y/y	mostly GPUs	end-June 2026	after the raises

Source: Nebius Group's own releases and shareholder letter. Convertible notes: \$2bn priced 10-Sep-2025 alongside a concurrent Class A share offering, and \$5.0bn priced 19-Aug-2026, upsized from \$4.5bn, in two tranches — \$3.0bn at 0.50% due 2030 and \$2.0bn at 4.50% due 2034, at conversion premiums of 40% and 45% over a \$223.90 share price, for net proceeds of about \$4.94bn. Secured debt: approximately \$775m announced 17-Jul-2026, maturing 31-Oct-2030 at SOFR plus 2.50%, arranged by MUFG with ABN AMRO, Bank of America, Deutsche Bank and HSBC as mandated lead arrangers. Meta agreement of up to approximately \$27bn over five years announced 16-Mar-2026, comprising \$12bn of dedicated capacity from early 2027 and up to \$15bn of further purchases; the Microsoft agreement of 8-Sep-2025 is described by Nebius only as multi-year and multi-billion dollar. Quarterly figures are Q2 2026 from the shareholder letter of 12-Aug-2026. The equity raised alongside the 2025 notes is not sized here because Nebius did not state it in that release.

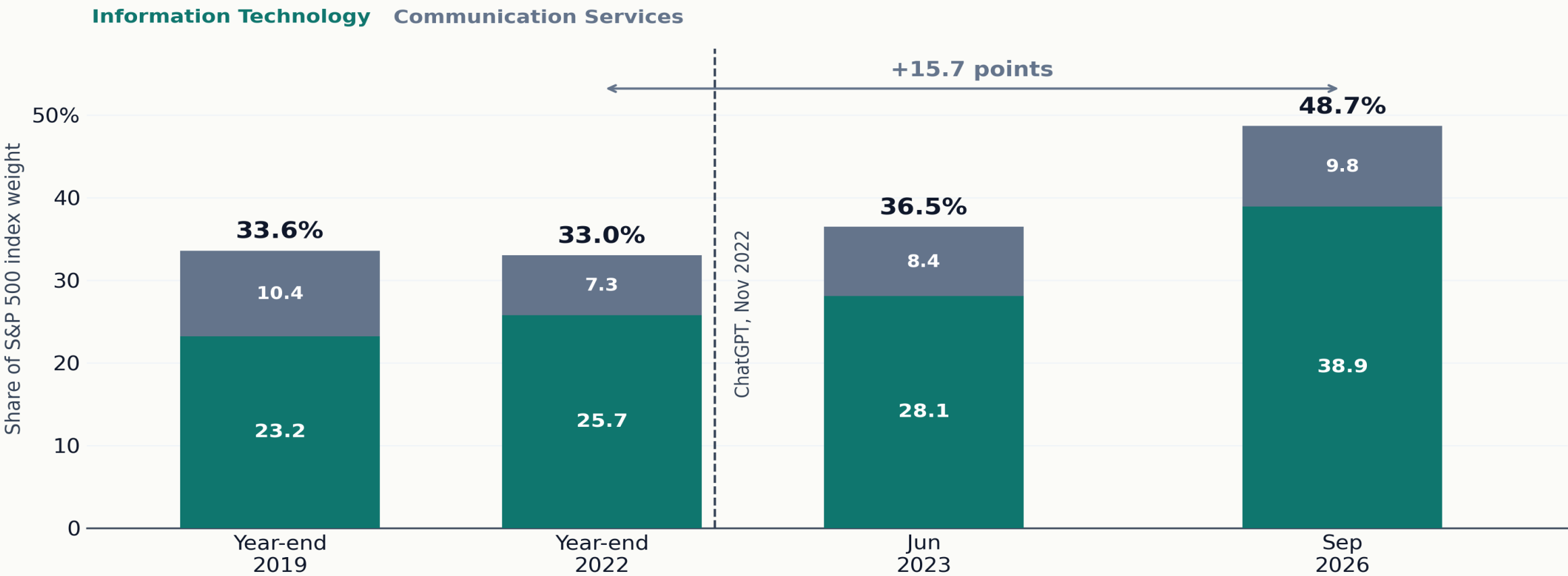
Equity markets fund the same story from both sides

- Public: Nvidia ~\$5.3T; the Magnificent 7 drove 42% of 2025's S&P 500 return.
- Private: OpenAI \$852B and Anthropic \$965B rounds closed in 2026.
- Over 70% of Q2 2026 global venture funding went to AI.

S&P Dow Jones Indices (Dec 2025); company announcements; Crunchbase (July 2026).

Exhibit 11: Tech and communications went from a third of the index to nearly half

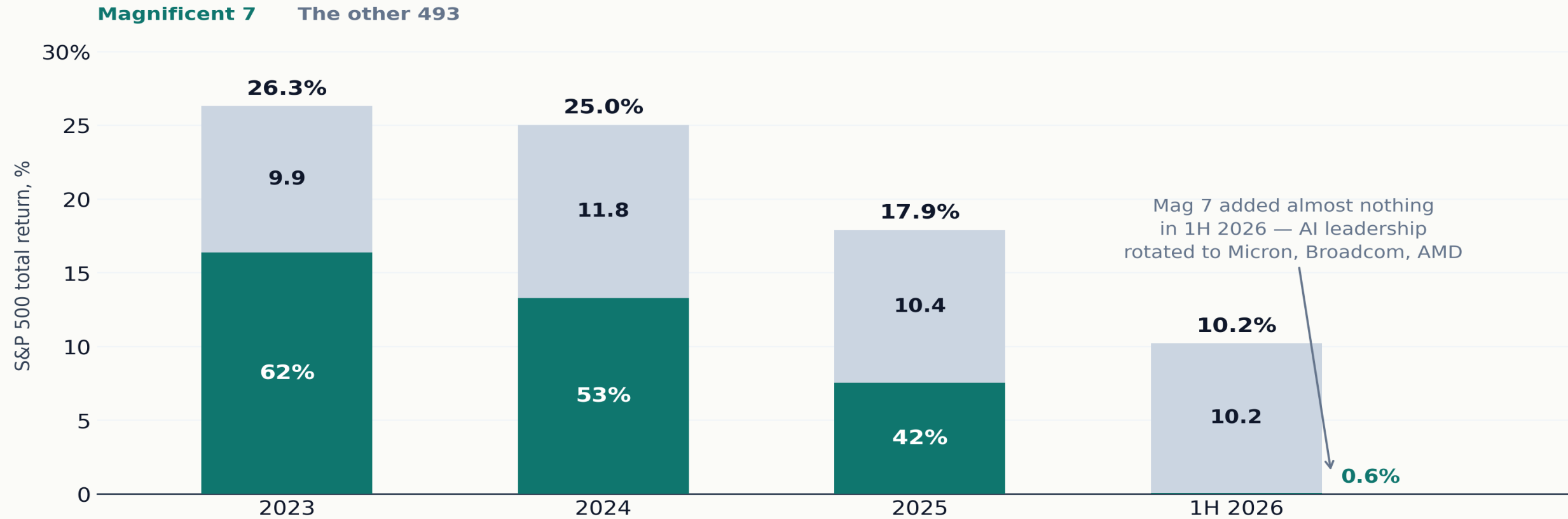
The combined weight was flat from 2019 to 2022 — then added about 15 points after ChatGPT.



Source: year-end 2019–2022 and Jun-2023 from Bespoke Investment Group, "S&P 500 Year End Sector Weightings Since 1990" (Jul 2023), GICS float-adjusted index weight. Sep-2026 is the iShares IVV holdings file (17 Sep 2026), corroborated by the SPDR SPY fact sheet at 30 Jun 2026 (47.7%) and ChartRow at 18 Sep 2026 (48.7%). Series starts in 2018 because GICS created Communication Services that September, moving Alphabet and Meta out of Info Tech — earlier years are not comparable. Year-end 2024 and 2025 readings could not be sourced, so these are dated markers, not a continuous line.

Exhibit 12: Seven stocks — a third of the index — drove 42-62% of its return

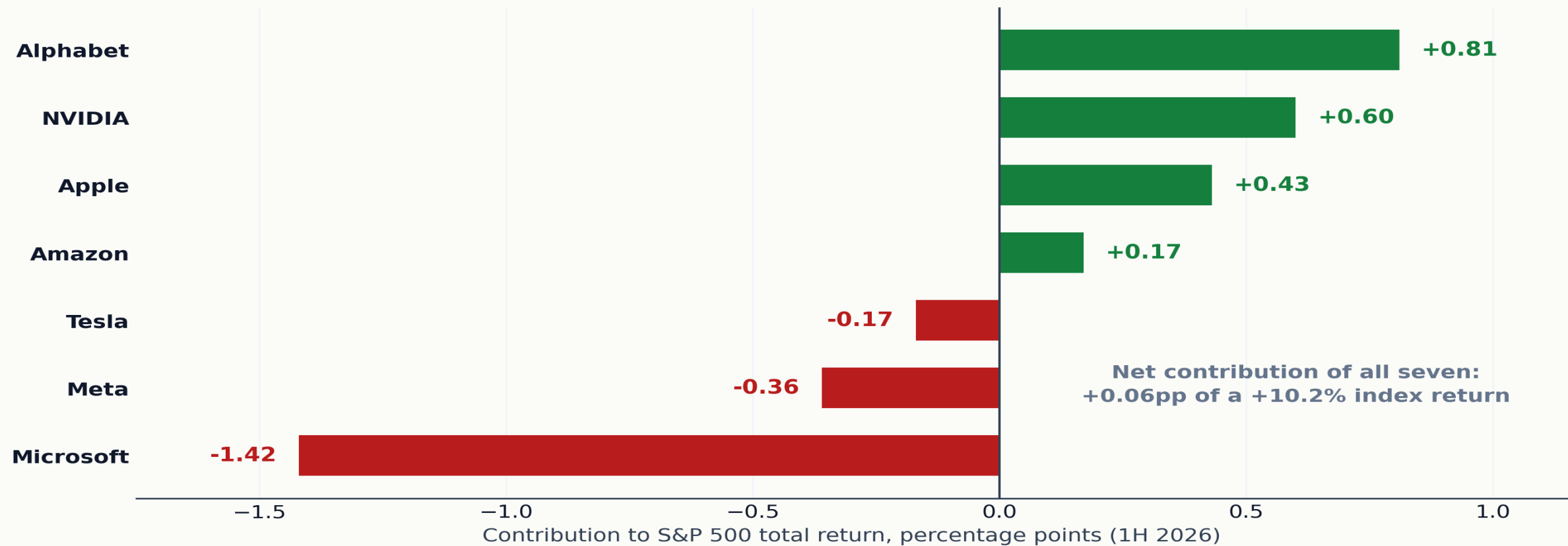
The other 493 delivered a steady ~10% a year. The Magnificent 7 supplied nearly all the variation.



Source: S&P Dow Jones Indices, Market Attributes: U.S. Equities (Dec 2023, Dec 2024, Dec 2025); First Trust (9 Jul 2026) for 1H 2026, same Magnificent 7 basket. Total return. Derived: 2023-24 splits are index return less S&P DJI's published ex-Mag-7 return; the 1H 2026 Mag 7 figure sums First Trust's seven published per-stock contributions. 2026 caveat: the Mag 7 lagged the index while AI leadership rotated to Micron, Broadcom, AMD and Intel — AI concentration persisted even as this basket stopped proxying it.

Exhibit 13: In 2026 the Magnificent 7 stopped moving together

Microsoft subtracted 1.4pp while Alphabet added 0.8pp. Together the seven contributed almost nothing.

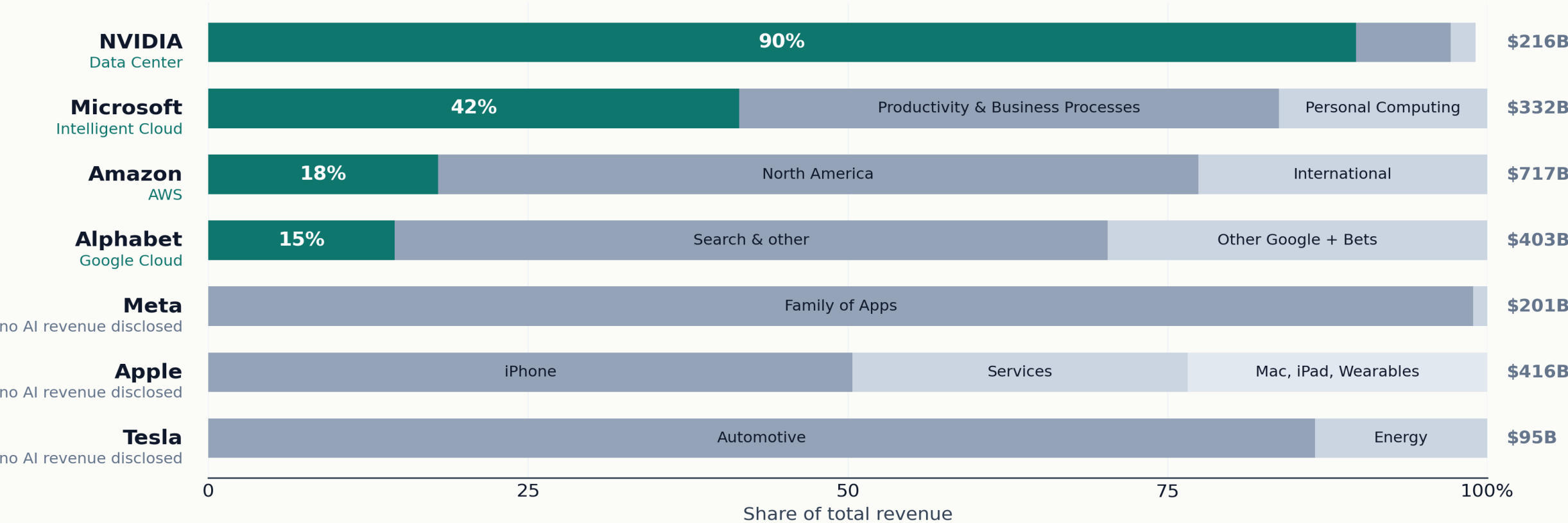


Source: First Trust, "S&P 500 Index 1H Update: The Broadening Continues" (9 Jul 2026), Bloomberg and Capital IQ data, 31 Dec 2025 – 30 Jun 2026. Contributions are percentage points of the index's +10.2% first-half total return; the net figure sums the seven published per-stock values. Context: State Street (20 Jul 2026, FactSet data) reports only three of the Magnificent 7 among the index's top 10 contributors in 2026, versus seven in 2024. AI leadership rotated to Micron, Broadcom, AMD and Intel.

Exhibit 14: Only NVIDIA earns most of its revenue from the AI-exposed segment

Six of the seven disclose no AI revenue at all. The rest still earn from phones, ads, retail and cars.

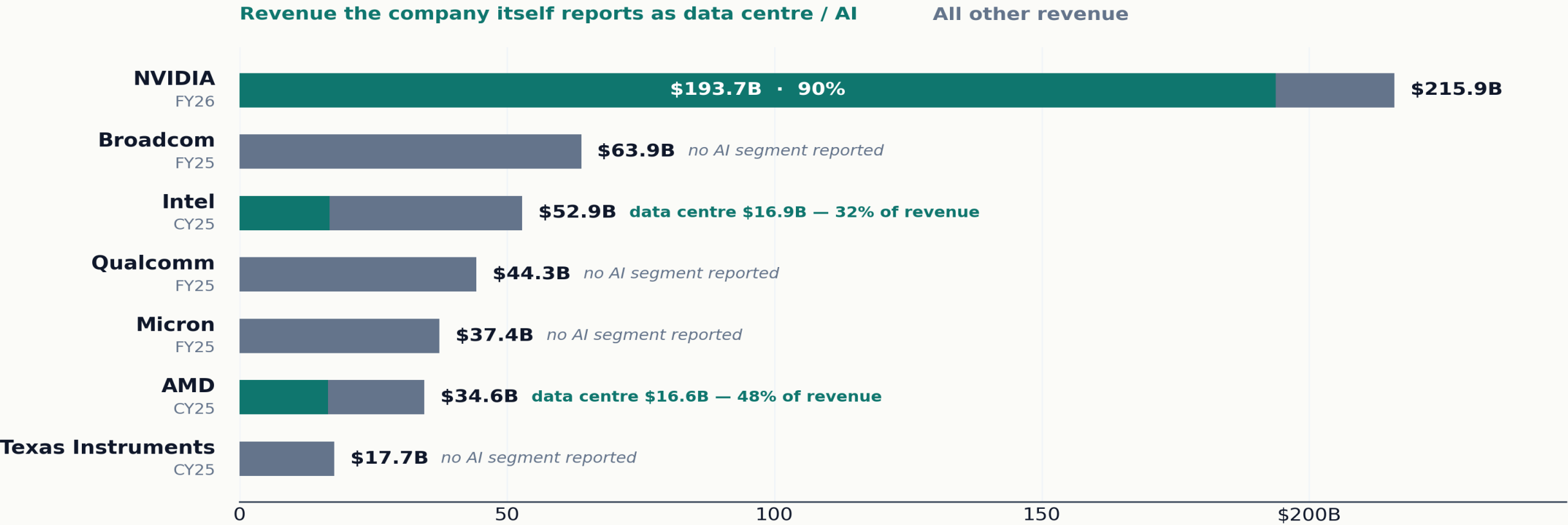
AI-exposed segment Everything else, largest first



Source: each company's most recent 10-K (SEC EDGAR XBRL) — Amazon, Alphabet, Meta and Tesla FY2025 (Dec-25); Apple FY2025 (Sep-25); NVIDIA FY2026 (Jan-26); Microsoft FY2026 (Jun-26). Segments are as filed; period-ends span nine months, so shares are comparable but the years are not coterminous. "AI-exposed" is this chart's label for the segment most tied to AI demand — no company reports a segment called AI. Microsoft alone publishes an AI figure: a \$37B annual run rate (FY26 Q3), a non-GAAP point-in-time metric outside the audited segment note, ~11% of revenue.

Exhibit 15: NVIDIA is not the biggest chip story — it is nearly the only one

Data centres are 90% of NVIDIA's revenue. Of the six other large US chipmakers, four disclose no full-year AI line at all.



Source: each company's own fourth-quarter and full-year earnings release, or the SEC 8-K exhibit thereto. NVIDIA FY2026 (ended Jan-26); Broadcom FY2025 (Nov-25); Qualcomm FY2025 (Sep-25); Micron FY2025 (Aug-25); Intel, AMD and Texas Instruments CY2025 (Dec-25). Period-ends span eleven months, so the ranking is indicative rather than like-for-like. The teal band is the segment the company itself reports as data centre or AI — NVIDIA "Data Center", Intel "Data Center and AI", AMD "Data Center". Broadcom, Qualcomm, Micron and TI publish no full-year figure on that basis: Broadcom reports AI semiconductor revenue only by quarter (Q4 FY25 +74% y/y; Q1 FY26 guided at \$8.2B) and Micron reports business-unit revenue only by quarter. Neither is summed here. TSMC, Samsung and SK hynix report in NT dollars and won and are excluded rather than converted at a rate this chart has not verified.

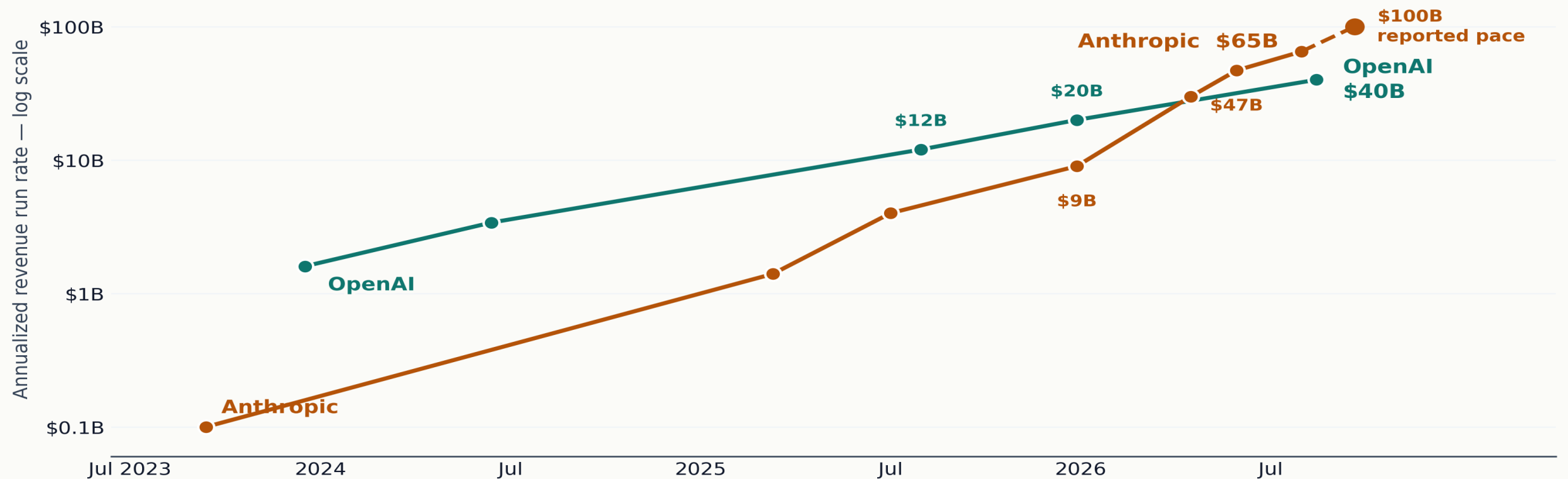
04

Science to business case

Model decisions used to be research bets. Now they are underwritten like any other capital project.

Exhibit 16: Anthropic's run rate overtook OpenAI's during 2026

Anthropic went from \$9B to \$65B in seven months. OpenAI roughly doubled to \$40B over the same stretch.



Every figure is an ANNUALIZED RUN RATE — a recent period multiplied up — not audited revenue. Neither company is listed and neither publishes audited revenue, so none of this is a filing. Company-stated: Anthropic's \$100M (2023, to investors), \$30B (Amodei, May 2026) and \$47B (Series H, May 2026); OpenAI's >\$20B (CFO, Jan 2026) and \$2B/month (Apr 2026). Press-reported from unnamed sources: OpenAI \$1.6B, \$3.4B, \$12B and \$40B (The Information, Reuters, Bloomberg); Anthropic \$1.4B, \$4B, \$9B and \$65B (The Information, Bloomberg). The \$100B is a forward pace reported by the New York Times in Sept 2026, not revenue earned over an elapsed period.

Collapsing inference prices are what turn models into products

- For constant quality, inference cost falls roughly 10x per year.
- GPT-3-level output: \$60 per million tokens (2021) → \$0.06 (2024).
- Cheap tokens make ads, agents, and enterprise software viable businesses.

a16z "LLMflation", Nov 2024.

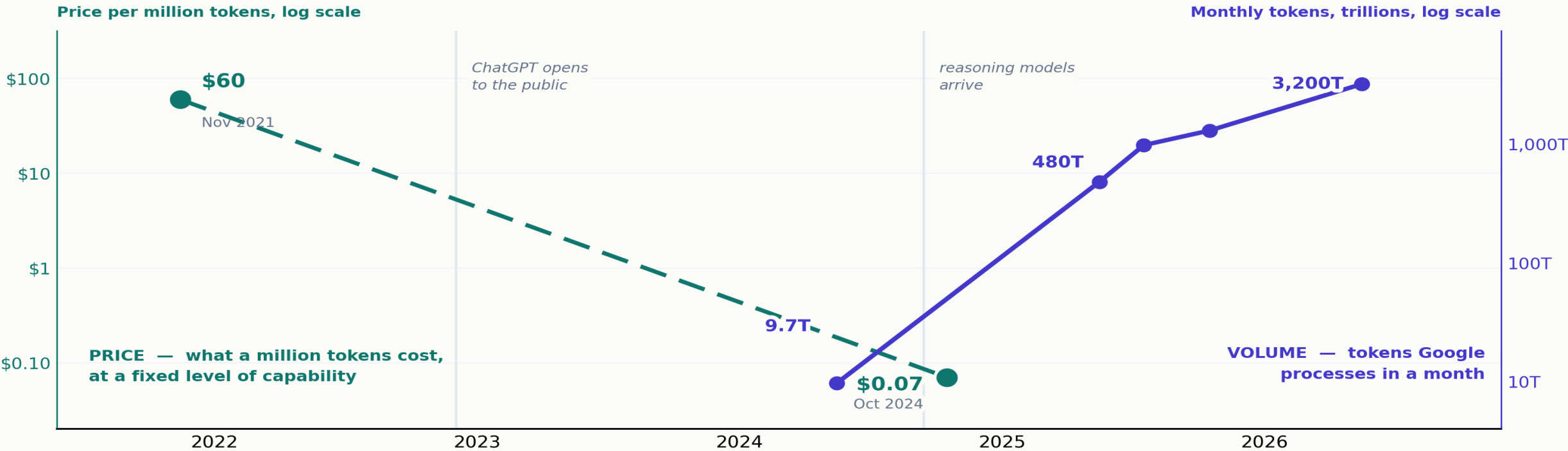
Spending is now justified by contracts, backlogs, and ROI

- Oracle's \$455B backlog, anchored by a \$300B OpenAI contract, backs its bonds.
- Vendor circularity: Nvidia invests in the labs that buy its chips.
- 88% of firms use AI, but few yet report earnings impact.

Analyst discipline: run rates ≠ GAAP revenue; commitments ≠ cash.

The price of a token fell about 900-fold. The number of tokens rose faster.

Cheap tokens do not shrink the compute bill. They open up uses that were uneconomic, and the volume more than replaces the price.



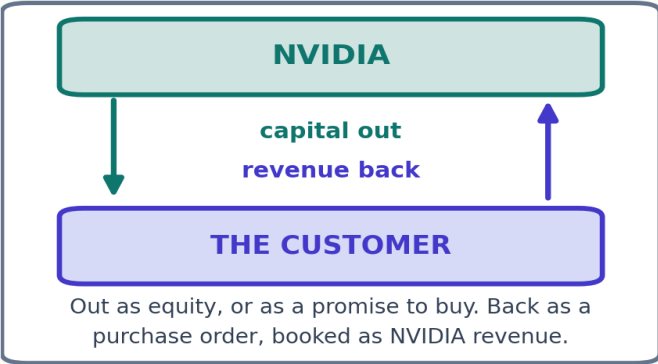
Each time the price falls, a use that was uneconomic becomes economic, and the tokens it consumes swamp the saving. Cheaper inference has not cut anyone's compute bill.

Sources. Price: Epoch AI, "LLM inference prices have fallen rapidly but unequally across tasks" — the price to reach GPT-3's level on MMLU fell from \$60 per million tokens in November 2021 to \$0.07 in October 2024. Epoch puts the fall for GPT-4-level performance on PhD-level science questions at 40x a year, and the range across the six milestones it tracks at 9x to 900x a year, cautioning that the fastest drops came in the final year so may not persist. Only Epoch's two dated endpoints are plotted; the line between them is the implied path, not a measured series, which is why it is dashed. Volume: Alphabet's own statements — 9.7 trillion monthly tokens in May 2024 and 480 trillion in May 2025 (Google I/O 2025), 980 trillion in July 2025 and over 1.3 quadrillion in the third quarter (CEO remarks with Q3 2025 earnings), and over 3.2 quadrillion in May 2026 (Google I/O 2026). The two series are not the same universe: the price line is Epoch's cross-vendor capability milestone, the volume line is one company's own disclosure. They are shown together to compare direction and slope, not levels, and no ratio between them is computed. Both axes are logarithmic, so a straight line is a constant rate of change, and neither series is adjusted for the mix of models or tasks behind it.

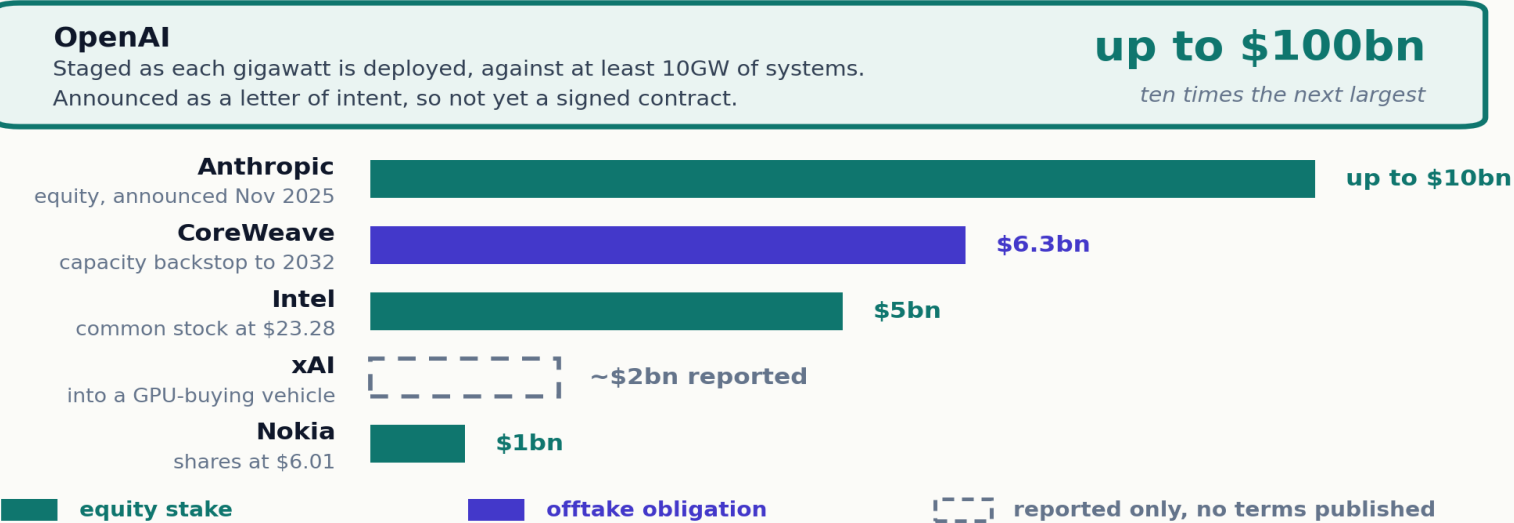
NVIDIA is funding several of the companies whose orders it books as revenue

Money goes out as equity or an offtake promise and returns as a purchase order. The loop is real — the six deals below are not one instrument.

THE LOOP



ANNOUNCED SIZE OF EACH COMMITMENT



These do not add up, and that is the point

A non-binding letter of intent staged over years, an obligation to buy unsold cloud capacity and an ordinary purchase of listed shares are three different things, so no total is shown. What they have in common is the direction of travel: NVIDIA supplies capital to a buyer, the buyer spends it on NVIDIA hardware, and the hardware is booked as NVIDIA revenue. Whether that is vendor financing, strategic investment or demand creation is the question to hold open — none of these releases says how much cash has actually moved, and none of the bars is money already deployed.

Sources, each the counterparty's own release or filing. OpenAI: NVIDIA newsroom, 22-Sep-2025 — "NVIDIA intends to invest up to \$100 billion in OpenAI progressively as each gigawatt is deployed", against at least 10GW of NVIDIA systems, announced as a letter of intent the parties "look forward to finalizing", so it is not a signed contract. Anthropic: Anthropic, Microsoft and NVIDIA, 18-Nov-2025 — NVIDIA and Microsoft "committing to invest up to \$10 billion and up to \$5 billion respectively", alongside Anthropic's commitment to purchase \$30bn of Azure capacity and up to a gigawatt of NVIDIA systems; only NVIDIA's leg is plotted. CoreWeave: CoreWeave Form 8-K filed for 9-Sep-2025 — an order form of \$6.3bn running through 13-Apr-2032 under which NVIDIA is obligated to purchase residual unsold capacity. That is an offtake obligation, not an investment, and is shown in a different colour for that reason. Intel: NVIDIA newsroom, 18-Sep-2025 — \$5bn of common stock at \$23.28 per share. Nokia: NVIDIA newsroom, 28-Oct-2025 — \$1bn at \$6.01 per share. xAI: reported by Bloomberg in October 2025 as roughly \$2bn into a \$20bn special-purpose vehicle that buys GPUs; neither company has published terms, so it is drawn hollow and dashed. The bars are announced headline sizes, not cash deployed, and they are not summed here: a staged non-binding letter of intent, a capacity backstop and an ordinary share purchase do not add.

What to take away

- 01 An LLM predicts the next token; training is capex, inference is opex — and both labs are building silicon to cut the second one.
- 02 Scaling laws made compute the input to progress — capex doubled into 2026.
- 03 Bonds, private credit, ABS, and mega-rounds now fund the buildout.
- 04 The frontier runs on business cases: revenue contracts, unit economics, and leverage.

Questions?

GO DEEPER

Epoch AI — training compute and cost trends (epoch.ai)

Morgan Stanley, "AI and the Credit Markets" (July 2025)

Stanford AI Index 2026 — hai.stanford.edu